Diarrhoeagenic *Escherichia coli* Disease Attribution and the Asymptomatic Colonization Paradox: A Systematic Review and Hierarchical Meta-Analysis

Abstract

Background

Diarrhoeagenic *Escherichia coli* (DEC) remains a primary bacterial cause of childhood morbidity, growth faltering, and mortality in Africa and globally. However, molecular surveillance studies frequently report high DEC prevalence in asymptomatic community controls, creating an epidemiological attribution paradox across enteroaggregative *E. coli* (EAEC), diffusely adherent *E. coli* (DAEC), and atypical enteropathogenic *E. coli* (aEPEC). Whether this paradox reflects true non-pathogenicity, diagnostic platform discordance, hybrid pathotype emergence, host age-dependent susceptibility, or unmeasured accessory virulence gene heterogeneity within canonical pathotypes remains unresolved. We conducted a systematic review and hierarchical random-effects meta-analysis following PRISMA 2020 guidelines to: (1) quantify the molecular prevalence, matched case-control odds ratios (ORs), diagnostic modality discordance, and hybrid pathotype architecture of DEC across African subregions and multicentre reference cohorts; and (2) evaluate locus-specific case-control pathogenicity across 29 genetic determinants in EAEC, DAEC, and EPEC in Africa and globally.

Methods

We searched PubMed/MEDLINE, Scopus, Web of Science, Embase, African Journals Online (AJOL), and NCBI/ENA genomic repositories through March 2026. At the pathotype level, the review synthesized $k = 28$ primary African and multicentre molecular and genomic cohorts ($N = 15,750$ participants: $9,360$ diarrhoeal cases and $6,041$ asymptomatic controls across 18 countries in all five United Nations African subregions), establishing the most comprehensive and methodologically rigorous evidence base of DEC molecular epidemiology compiled to date. At the determinant level, the review synthesized $k = 30$ global and African case-control cohorts (154 locus-level 2×2 evaluations across Africa, Latin America, Asia, Europe, and Oceania). Single-arm proportions were pooled using the random-effects meta-analysis models to pool proportions and 95% confidence intervals across cohorts. Case-control associations were pooled as log Odds Ratios ($\ln \mathrm{OR}$) using REML and random-effects (`DL`) estimators alongside small-sample adjusted small-sample t-distribution adjustments, I^2 and τ^2 heterogeneity metrics, and Egger's regression and Begg's rank-correlation publication bias tests.

Results

At the pathotype level, pooled random-effects prevalence among African diarrhoeal cases was highest for EAEC (28.3%, 95% CI: 23.2–33.7%; $k = 26$, $2,406 / 9,264$), followed by EPEC (16.1%, 95% CI: 12.6–19.8%; $k = 26$, $1,497 / 9,264$), enterotoxigenic *E. coli* (ETEC; 14.9%, 95% CI: 11.8–18.1%; $k = 26$, $1,426 / 9,264$), atypical EPEC (10.9%, 95% CI: 6.5–16.3%; $k = 11$), typical EPEC (6.0%, 95% CI: 4.2–8.1%; $k = 12$), DAEC (4.9%, 95% CI: 3.8–6.2%;

$k = 22$), Shiga toxin-producing *E. coli* (STEC; 3.2 %, 95 % CI: 2.0 – 4.7 %; $k = 25$), and enteroinvasive *E. coli* (EIEC; 1.4 %, 95 % CI: 1.0 – 1.9 %; $k = 26$). In matched case-control analyses, acute diarrhoeal disease was strongly associated with EIEC (OR = 2.43, 95 % CI: 1.69 – 3.50, $P < 0.0001$), tEPEC (OR = 2.42, 95 % CI: 1.96 – 2.99, $P < 0.0001$), and ETEC (OR = 2.37, 95 % CI: 2.08 – 2.70, $P < 0.0001$), whereas unstratified EAEC (OR = 1.38, 95 % CI: 1.03 – 1.85, $P = 0.0326$; KH 95 % CI: 0.95 – 2.00), unstratified EPEC (OR = 1.20, 95 % CI: 1.08 – 1.35, $P = 0.0011$), unstratified DAEC (OR = 1.21, 95 % CI: 0.86 – 1.71, $P = 0.2689$), and unstratified aEPEC (OR = 0.89, 95 % CI: 0.78 – 1.01, $P = 0.0694$) exhibited attenuated or null associations due to widespread asymptomatic carriage. Diagnostic platform stratification revealed substantial sensitivity gradients: EAEC detection rose from 23.5 % (95 % CI: 19.1 – 28.3 %) by conventional endpoint PCR to 39.2 % (95 % CI: 24.6 – 54.9 %) by quantitative real-time PCR / TaqMan Array Card (qPCR/TAC) and 41.1 % (95 % CI: 36.9 – 45.5 %) by whole-genome sequencing (WGS). Hybrid pathotypes (`DEC–DEC` and `DEC–ExPEC`) accounted for 7.4 % (95 % CI: 5.5 – 9.6 %; $k = 27,502 / 9,039$) of DEC-positive specimens overall, rising from 5.5 % by conventional PCR to 13.9 % (95 % CI: 10.7 – 17.5 %) under WGS, where conventional endpoint PCR exhibited a 22.2 % (41 / 185) diagnostic discordance rate against WGS reference assemblies. At the determinant level, locus-resolved synthesis across African and global case-control cohorts resolved the carriage paradox across all three pathotypes: (1) within EAEC, plasmid-encoded `aatA` (OR = 1.21, 95 % CI: 1.01 – 1.45), `aggR` (OR = 1.17, 95 % CI: 0.98 – 1.40), `aap` (OR = 1.09, $P = 0.53$), `pic` (OR = 1.04, 95 % CI: 0.79 – 1.37, $P = 0.77$, calibrated literature prevalence 16 – 22 %), `astA` (OR = 1.06, $P = 0.59$), and `irp2/fyuA` (OR = 1.14, $P = 0.49$) did not strongly discriminate cases from controls, whereas the SPATE protease `sepA` (OR = 4.68, 95 % CI: 2.85 – 7.69, $P = 1.15 \times 10^{-9}$, $I^2 = 0.0$ %), colonization factor `CS22` (OR = 3.59, 95 % CI: 1.44 – 8.94, $P = 0.0061$), AAF/II fimbrial subunit `aafA` (OR = 3.08, 95 % CI: 2.00 – 4.74, $P = 3.25 \times 10^{-7}$), AAF/III subunit `agg3A` (OR = 2.71, 95 % CI: 1.28 – 5.74, $P = 0.0090$), combined AAF/I–V fimbriae (OR = 2.38, 95 % CI: 1.68 – 3.37, $P = 1.06 \times 10^{-6}$), cytotoxic SPATE `pet` (OR = 2.19, 95 % CI: 1.30 – 3.69, $P = 0.0032$), vacuolating SPATE `sat` (OR = 1.78, 95 % CI: 1.24 – 2.56, $P = 0.0018$), and chromosomal T6SS `aaiC` (OR = 1.47, 95 % CI: 1.11 – 1.96, $P = 0.0076$) were significantly enriched in diarrhoeal cases; (2) within DAEC (`daaC` / `afa/draBC`), pathogenicity displayed a sharp host age-ontogeny switch, shifting from null association in infants aged < 12 months (OR = 0.91, 95 % CI: 0.69 – 1.20, $P = 0.5094$, $I^2 = 0.0$ %) to strong pathogenic attribution in children aged ≥ 12 – 59 months (OR = 2.21, 95 % CI: 1.73 – 2.82, $P = 1.84 \times 10^{-10}$, $I^2 = 0.0$ %), amplified by `sat` (OR = 3.22, 95 % CI: 1.72 – 6.02, $P = 0.0002$) and `afaE` subtypes I/III/V (OR = 3.07, 95 % CI: 1.75 – 5.41, $P < 0.0001$); and (3) within EPEC, `bfpA+` tEPEC was strongly pathogenic in infants < 12 months (OR = 2.82, 95 % CI: 2.27 – 3.52, $P < 10^{-10}$) and young children ≥ 12 – 59 months (OR = 1.89, 95 % CI: 1.45 – 2.47, $P = 2.29 \times 10^{-6}$), whereas unstratified `eae+ bfpA-` aEPEC was neutral (OR = 1.06, 95 % CI: 0.85 – 1.32, $P = 0.6176$) unless carrying the OI-122 pathogenicity island effectors `lifA/efa1` (OR = 6.01, 95 % CI: 2.64 – 13.69, $P = 1.94 \times 10^{-5}$), `nleB` (OR = 4.20, 95 % CI: 2.06 – 8.56, $P = 7.61 \times 10^{-5}$), `nleE` (OR = 3.80, 95 % CI: 1.86 – 7.75, $P = 0.0002$), or `astA` (OR = 2.63, 95 % CI: 1.28 – 5.37, $P = 0.0082$).

Conclusions

Pathotype-wide molecular screening for `aatA`/`aggR`, unstratified `eae`, or age-unstratified `daaC`/`afaBC` conflates high-burden asymptomatic colonization with virulent infection in Africa

and globally. Clinical attribution and molecular surveillance panels should incorporate second-tier effector and adhesin targets (`sepA`, `aafA`, `agg3A`, `CS22`, `pet`, `sat`, `bfpA`, and OI-122 `lifA/efa1`–`nleB`–`nleE`) alongside age-stratified interpretation for DAEC (≥12 months) and multi-target screening capable of resolving two-way hybrid DEC lineages.

Keywords: Diarrhoeagenic *Escherichia coli*; Enteroaggregative *Escherichia coli*; Diffusely Adherent *Escherichia coli*; Enteropathogenic *Escherichia coli*; Hybrid Pathotypes; Virulence Loci; Diagnostic Modality; Africa; Global Health; Systematic Review; Meta-Analysis.

Author Summary

Why Was This Study Done?

- Diarrhoeagenic *Escherichia coli* (DEC) is a leading bacterial cause of childhood diarrhoea, intestinal inflammation, and linear growth faltering across Africa and globally.
- Paradoxically, three of the most frequently detected DEC pathotypes—enteroaggregative *E. coli* (EAEC), atypical enteropathogenic *E. coli* (aEPEC), and diffusely adherent *E. coli* (DAEC)—are often recovered at similarly high rates from both children with acute diarrhoea and asymptomatic community controls when screened using standard single-gene PCR targets (`aatA`, `aggR`, `eae`, or `daaC`).
- Clinical microbiologists and infectious disease epidemiologists have lacked a unified quantitative synthesis determining whether this attribution paradox is driven by diagnostic platform sensitivity (endpoint PCR vs. quantitative PCR vs. whole-genome sequencing [WGS]), hybrid pathotype emergence, host developmental age, or specific accessory virulence genes within each pathotype.

What Did the Researchers Do and Find?

- We conducted a systematic review and hierarchical random-effects meta-analysis following PRISMA 2020 guidelines synthesizing $k = 28$ primary African and multicentre molecular/genomic studies ($N = 15,750$ participants) across 18 African nations at the pathotype level, and $k = 30$ global and African case-control cohorts (154 locus-level 2×2 evaluations across five continents) at the determinant level.
- In African surveillance cohorts, EAEC (28.3%), EPEC (16.1%), and ETEC (14.9%) dominated diarrhoeal cases, but matched case-control pathogenicity was high only for EIEC (OR = 2.43), typical EPEC (OR = 2.42), and ETEC (OR = 2.37), whereas unstratified EAEC (OR = 1.38), EPEC (OR = 1.20), DAEC (OR = 1.21), and aEPEC (OR = 0.89) showed weak or null associations.
- Transitioning from conventional endpoint PCR to qPCR/TAC and WGS nearly doubled EAEC detection (23.5% to 41.1%) and more than doubled hybrid pathotype identification (5.5% to 13.9%), exposing a 22.2% endpoint PCR false-negative/misclassification rate relative to WGS.
- Locus-level global and African synthesis resolved the carriage paradox: acute clinical pathogenicity is driven by specific accessory effectors—`sepA` (OR = 4.68), `CS22` (OR = 3.59), `aafA` (OR = 3.08), `agg3A` (OR = 2.71), `pet` (OR = 2.19), `sat` (OR = 1.78), and `aaiC` (OR = 1.47) in EAEC; host age ≥12–59 months (OR = 2.21 vs. OR = 0.91 in

infants < 12 months), `sat` (OR = 3.22), and `afaE` (OR = 3.07) in DAEC; and `bfpA` (OR = 2.82 in infants) or the OI-122 island effectors `lifA/efa1` (OR = 6.01), `nleB` (OR = 4.20), and `nleE` (OR = 3.80) in aEPEC.

What Do These Findings Mean?

- Routine diagnostic panels that rely solely on canonical backbone markers (`aatA`/`aggR` for EAEC, unstratified `eae` for aEPEC, or age-unstratified `daaC`/`afaBC` for DAEC) cannot distinguish widespread endemic colonization from virulent infection.
- Clinical microbiology laboratories, enteric surveillance networks, and vaccine developers in Africa and globally should incorporate second-tier adhesin and toxin targets (`sepA`, `aafA`, `agg3A`, `CS22`, `pet`, `sat`, `bfpA`, and OI-122 `lifA/efa1`–`nleB`–`nleE`), apply age-stratified interpretation for DAEC in children aged ≥ 12 months, and adopt multi-locus molecular or genomic workflows capable of detecting two-way hybrid DEC lineages.

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1. Introduction

1.1 Global and African Burden of Diarrhoeagenic *Escherichia coli* (DEC)

Diarrhoeal disease remains a leading cause of preventable morbidity, linear growth faltering, and under-five mortality worldwide, with the heaviest clinical burden concentrated in sub-Saharan Africa and South Asia (Kotloff et al., 2013; Platts-Mills et al., 2015; Troeger et al., 2018). Among bacterial enteropathogens, diarrhoeagenic *Escherichia coli* (DEC) represents the most common and genetically diverse group colonizing and infecting infants and young children in low- and middle-income countries (LMICs) (Nataro & Kaper, 1998; Croxen et al., 2013; Kaper et al., 2004). Since the classic synthesis by Nataro and Kaper (1998), DEC isolates have been classified into six canonical pathotypes based on conserved virulence plasmids, chromosomal pathogenicity islands, and in vitro tissue-culture adherence phenotypes: 1. **Enteropathogenic *E. coli* (EPEC):** Characterized by the chromosomal Locus of Enterocyte Effacement (LEE) encoding the outer-membrane adhesin intimin (`eae`) that induces attaching-and-effacing (A/E) lesions, further subdivided by the presence (`tEPEC`) or absence (`aEPEC`) of the EPEC adherence factor (EAF) plasmid encoding the bundle-forming pilus (`bfpA`) (Kaper et al., 2004; Trabulsi et al., 2002). 2. **Enterotoxigenic *E. coli* (ETEC):** Defined by plasmid-encoded heat-labile (`elt` / LT) and/or heat-stable (`estA` / STp, STh) enterotoxins paired with over 25 antigenically distinct colonization factors (CFs) that mediate small-intestinal attachment and secretory diarrhoea (Qadri et al., 2005; Kotloff et al., 2013). 3. **Enteroaggregative *E. coli* (EAEC):** Phenotypically defined by stacked-brick aggregative adherence to HEp-2 cells and molecularly identified by the pAA virulence plasmid encoding the AraC/XylS-family master transcriptional regulator `aggR`, the anti-aggregation protein transporter `aatA` (CVD432 probe), dispersin (`aap`), aggregative adherence fimbriae (`AAF/I–V`), and chromosomal type VI secretion systems (`aaiC`) (Harrington et al., 2006; Estrada-Garcia & Navarro-Garcia, 2012; Hebbelstrup Jensen et al., 2014). 4. **Shiga toxin-producing / Enterohaemorrhagic *E. coli* (STEC/EHEC):** Defined by lambdoid bacteriophage-encoded Shiga toxins (`stx1`, `stx2`), with the EHEC subset additionally harbouring the LEE island (`eae`) and enterohaemolysin (`ehxA`) associated with haemorrhagic colitis and haemolytic uraemic syndrome (Kaper et al., 2004; Lupindu, 2018). 5. **Enteroinvasive *E. coli* (EIEC):** Closely related to *Shigella* spp., harbouring the large invasion plasmid pINV (`ipaH`, `ial`, `virF`) that directs colonic epithelial invasion and dysentery (Croxen et al., 2013). 6.

Diffusely Adherent *E. coli* (DAEC): Characterized by diffuse adherence across the entire HEp-2 cell surface mediated by Afa/Dr diffusely adhering fimbrial and afimbrial adhesins (`afa/draBC`, `daaC`, `daaE`, `afaE` subtypes) that bind human decay-accelerating factor (DAF/CD55) and carcinoembryonic antigen-related cell adhesion molecules (CEACAMs) (Servin, 2005;Servin, 2014).

1.2 The Epidemiological Attribution Paradox in Africa and Globally

Over the past two decades, landmark multicentre studies—including the seven-site Global Enteric Multicenter Study (GEMS) across four African countries (The Gambia, Mali, Mozambique, and Kenya) and three Asian sites (Kotloff et al., 2013; Liu et al., 2016), the eight-country Malnutrition and Enteric Disease Study (MAL-ED) (Platts-Mills et al., 2015), and the Vaccine Impact on Diarrhea in Africa (VIDA) study in The Gambia, Mali, and Kenya (Kasumba et al., 2023; Ochieng et al., 2023)—alongside single-country investigations in Nigeria (Okeke et al., 2000; Onanuga et al., 2014; Saka et al., 2019; Akinlabi et al., 2023a), Mozambique (Mandomando et al., 2007; Garrine et al., 2020; Manhique-Coutinho et al., 2022), Kenya (Sang et al., 2012; Shah et al., 2016), South Africa (Tau et al., 2012; Tau et al., 2012), Ghana (Opintan et al., 2010), Tanzania (Moyo et al., 2007; Vargas et al., 2004), and Latin America (Girón et al., 1991; Levine et al., 1993; Scaletsky et al., 2002; Ochoa et al., 2009; Lima et al., 2013) have documented a striking epidemiological paradox.

When single-arm case prevalence is measured in children presenting with diarrhoea, **EAEC** (20 – 45 %), **aEPEC** (8 – 20 %), and **DAEC** (4 – 15 %) consistently rank among the most frequently detected bacterial enteropathogens in Africa and globally (Kalule et al., 2024; Lima et al., 2013). Yet when matched asymptomatic community controls are simultaneously enrolled, the unstratified case-control odds ratios for EAEC (`aatA`/`aggR`), aEPEC (`eae+ bfpA-`), and overall DAEC (`daaC`/`afaBC`) frequently hover near unity (OR ≈ 0.85 – 1.38), contrasting sharply with the robust case-control pathogenicity of ETEC (OR > 2.0), tEPEC (OR > 2.0), and EIEC (OR > 2.0) (Kotloff et al., 2013; Platts-Mills et al., 2015; Akinlabi et al., 2023a). Even in the absence of overt acute watery diarrhoea, subclinical colonization with EAEC and aEPEC is strongly associated with intestinal inflammation (elevated faecal myeloperoxidase, calprotectin, and neopterin), environmental enteric dysfunction (EED), and linear growth stunting across African and Latin American birth cohorts (Lima et al., 2013; Rogawski et al., 2017; Havt et al., 2017).

1.3 Four Competing Hypotheses Underlying the Attribution Paradox

Four non-mutually exclusive hypotheses have been proposed in the global literature to explain the divergence between high single-arm prevalence and attenuated case-control attribution: 1. **Diagnostic Modality Sensitivity and Primer Discordance:** Conventional endpoint multiplex PCR assays targeting conserved plasmid markers suffer from primer-binding mismatches in diverse African lineages and fail to quantify pathogen load, whereas quantitative real-time PCR (qPCR / TaqMan Array Card [TAC]) and whole-genome sequencing (WGS) reveal substantially higher pathogen prevalence, co-infection rates, and allelic diversity (Liu et al., 2016; Platts-Mills et al., 2015; Akinlabi et al., 2023b). 2. **Emergence of Two-Way Hybrid Pathotypes (`DEC–DEC` and `DEC–ExPEC`):** Horizontal transfer of conjugative virulence plasmids and bacteriophages across *E. coli* phylogroups generates hybrid pathotypes—such as `EAEC/ETEC`, `EAEC/EPEC`, `EAEC/STEC`, `EPEC/ETEC`, and `EAEC/UPEC`—that combine the mucosal biofilm persistence of

EAEC or ExPEC with acute enterotoxins or cytotoxins, confounding rigid single-marker diagnostic schemes (Karch et al., 2012; Santos et al., 2020; Akinlabi et al., 2025; Medugu et al., 2025). 3. **Host Developmental Age-Ontogeny:** Mucosal receptor expression (such as brush-border DAF/CD55 and CEACAMs bound by Afa/Dr DAEC adhesins) and passive maternal secretory IgA immunity against plasmid-encoded fimbriae evolve rapidly across the first two years of life, potentially rendering a pathotype non-pathogenic in young infants (<12 months) while driving acute diarrhoea in weaned toddlers ($\geq 12-59$ months) (Levine et al., 1993; Scaletsky et al., 2002; Ochoa et al., 2009; Garrine et al., 2020). 4. **Accessory Virulence Locus Heterogeneity Within Canonical Pathotypes:** Canonical pathotype-defining markers (`aatA`, `aggR`, `eae`, `daaC`) capture broad genetic backbones that include both avirulent commensal-like strains and hypervirulent clinical clones harbouring specific Serine Protease Autotransporters of *Enterobacteriaceae* (SPATEs; `sepA`, `pet`, `sat`), fimbrial adhesin variants (`aafA` [AAF/II], `agg3A` [AAF/III], `CS22`, `afaE-I/III/V`), or non-LEE-encoded type III secretion system effectors on genomic island OI-122 (`lifA/efa1`, `nleB`, `nleE`) (Boisen et al., 2012; Boisen et al., 2020; Afset et al., 2006; Hazen et al., 2016; Ochieng et al., 2023).

1.4 Rationale and Objectives of the Hierarchical Meta-Analysis

While earlier regional reviews (e.g., Kalule et al., 2024) provided valuable single-arm prevalence compilations across African hospital series through 2022, critical epidemiological, diagnostic, and genetic enigmas have remained unaddressed. Specifically, no quantitative meta-analysis has simultaneously: (i) updated the African evidence base through March 2026 to incorporate recent large-scale WGS and TAC cohorts ($k = 28$ primary studies, $N = 15,750$); (ii) directly contrasted single-arm Freeman–Tukey random-effects case prevalence against matched case-control odds ratios across all DEC pathotypes; (iii) quantified diagnostic discordance and two-way hybrid pathotype detection across conventional PCR, qPCR/TAC, and WGS platforms; and (iv) synthesized the global and African case-control literature at the individual virulence gene level ($k = 30$ cohorts, 154 locus-level 2×2 evaluations across five continents) to identify which specific genetic determinants and host age strata resolve the EAEC, DAEC, and EPEC carriage paradox.

Here, we report a comprehensive, systematic review and hierarchical random-effects meta-analysis following PRISMA 2020 guidelines addressing DEC pathotypes, diagnostic modalities, hybrid lineages, and virulence loci in Africa and globally.

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2. Methods

2.1 Study Design and Reporting Framework

This investigation was designed and executed as a hierarchical quantitative systematic review and meta-analysis in strict accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) statement (Page et al., 2021) (**Supplementary Table S1**) and the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) guidelines. The dual-track workflow is illustrated in **Figure 1**: - **Phase 1 (African & Multicentre DEC Pathotype Epidemiology, Diagnostic Platforms, and Hybrid Lineages):** Synthesizes primary molecular and genomic studies conducted across all five United Nations African subregions (Western, Eastern, Southern, Central, and Northern Africa), alongside

multicentre reference cohorts with substantial African enrolment (GEMS, MAL-ED, and VIDA), published between January 1, 2000, and March 25, 2026, establishing an updated, high-resolution continental evidence base across 18 countries. - **Phase 2 (Global & African Virulence Locus Case-Control Meta-Analysis):** Synthesizes matched and unmatched case-control studies from Africa, Latin America, Asia, Europe, and Oceania (January 1, 1990 – March 25, 2026) reporting extractable 2 × 2 contingency counts (cases positive/total vs. asymptomatic controls positive/total) for individual virulence loci within EAEC (`sepA`, `CS22`, `aafA`, `agg3A`, `agg4A`, `aggA`, `AAF_any`, `pet`, `sat`, `aaiC`, `aatA`, `aggR`, `irp2`, `fyuA`, `aap`, `pic`, `astA`), DAEC (`daaC`/`afaBC` overall and stratified by `<12` vs. `>=12–59` months, `afaE`, and `sat`), and EPEC (`bfpA` stratified by `<12` vs. `>=12–59` months, unstratified `eae`, and OI-122 effectors `lifA/efa1`, `nleB`, `nleE`, and `astA`).

2.2 Information Sources and Search Strategy

Systematic literature searches were executed across six bibliographic databases and genomic repositories: PubMed/MEDLINE, Scopus, Web of Science Core Collection (WoS), Embase, EBSCOhost, and African Journals Online (AJOL), supplemented by targeted screening of NCBI Sequence Read Archive (SRA) / European Nucleotide Archive (ENA) BioProjects and reference lists of prior reviews (Hebbelstrup Jensen et al., 2014; Lupindu, 2018; Kalule et al., 2024).

For **the African surveillance synthesis**, the Boolean search string combined pathotype terms, diagnostic terms, and geographic filters for all 54 African sovereign states: > `("diarrhoeagenic Escherichia coli" OR "diarrheagenic Escherichia coli" OR "DEC" OR "enteroaggregative" OR "EAEC" OR "enteropathogenic" OR "EPEC" OR "enterotoxigenic" OR "ETEC" OR "enteroinvasive" OR "EIEC" OR "Shiga toxin-producing" OR "STEC" OR "enterohaemorrhagic" OR "EHEC" OR "diffusely adherent" OR "DAEC" OR "hybrid Escherichia coli" OR "heteropathogenic") AND ("PCR" OR "multiplex PCR" OR "qPCR" OR "TaqMan Array Card" OR "whole genome sequencing" OR "WGS" OR "virulence genes") AND ("Africa" OR "sub-Saharan Africa" OR "West Africa" OR "East Africa" OR "Southern Africa" OR "Central Africa" OR "North Africa" OR "Algeria" OR "Angola" OR "Benin" OR "Botswana" OR "Burkina Faso" OR "Burundi" OR "Cabo Verde" OR "Cameroon" OR "Central African Republic" OR "Chad" OR "Comoros" OR "Congo" OR "Cote d'Ivoire" OR "Djibouti" OR "Egypt" OR "Equatorial Guinea" OR "Eritrea" OR "Eswatini" OR "Ethiopia" OR "Gabon" OR "Gambia" OR "Ghana" OR "Guinea" OR "Guinea-Bissau" OR "Kenya" OR "Lesotho" OR "Liberia" OR "Libya" OR "Madagascar" OR "Malawi" OR "Mali" OR "Mauritania" OR "Mauritius" OR "Morocco" OR "Mozambique" OR "Namibia" OR "Niger" OR "Nigeria" OR "Rwanda" OR "Sao Tome" OR "Senegal" OR "Seychelles" OR "Sierra Leone" OR "Somalia" OR "South Africa" OR "South Sudan" OR "Sudan" OR "Tanzania" OR "Togo" OR "Tunisia" OR "Uganda" OR "Zambia" OR "Zimbabwe")`.

For **the virulome attribution synthesis**, the global search removed geographic restrictions and targeted locus-specific case-control attribution: > `("enteroaggregative Escherichia coli" OR "EAEC" OR "diffusely adherent Escherichia coli" OR "DAEC" OR "enteropathogenic Escherichia coli" OR "EPEC" OR "atypical EPEC" OR "aEPEC") AND ("case-control" OR "healthy controls" OR "asymptomatic`

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controls" OR "diarrhea" OR "diarrhoea") AND ("aggR" OR "aatA" OR
"CVD432" OR "aap" OR "aaiC" OR "sepA" OR "pet" OR "sat" OR "pic" OR
"astA" OR "aafA" OR "aggA" OR "agg3A" OR "agg4A" OR "CS22" OR "irp2"
OR "fyuA" OR "daaC" OR "afaBC" OR "draBC" OR "afaE" OR "bfpA" OR "eae"
OR "lifA" OR "efa1" OR "nleB" OR "nleE" OR "OI-122").
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2.3 Eligibility Criteria (PICOS Framework)

1. **Population:** Human patients presenting with acute or persistent diarrhoea (cases) and, where applicable, asymptomatic community or hospital controls without diarrhoea for at least 7–14 days prior to sampling. Studies restricted exclusively to animal reservoirs, food matrices, or environmental water without human clinical isolates were excluded.
2. **Intervention / Exposure (Diagnostic Ascertainment):** Molecular confirmation of DEC pathotypes or individual virulence genes via endpoint PCR, multiplex PCR, quantitative real-time PCR (qPCR / TAC), DNA hybridization probes (`CVD432`, `daaC`), or whole-genome sequencing (WGS). Studies relying solely on O:H serotyping or biochemical profiling without molecular/probe confirmation were excluded (Okeke et al., 2000; Kalule et al., 2024).
3. **Comparator:** For case-control odds ratio models in the pathotype and locus-level analyses, an explicitly defined asymptomatic human control group from the same geographic source population tested using the identical diagnostic platform.
4. **Outcomes:** Extractable numerator and denominator counts (r / n) for pathotype-level case prevalence , matched case-control pathotype counts (a , b , c , d; pathotype synthesis), hybrid pathotype co-detections , and locus-specific 2×2 case-control counts across EAEC, DAEC, and EPEC .
5. **Handling of Unassayed Pathotypes and Overlapping Cohorts:** When a primary study assayed only a subset of DEC pathotypes (e.g., EAEC-specific WGS cohorts or EPEC/ETEC-focused surveys), unassayed pathotypes were coded as missing (`NA` / `-1`) rather than zero so that pathotype-specific random-effects denominators included only cohorts that actively tested for that pathotype. Where multiple publications analysed overlapping subsets of the same parent cohort (e.g., GEMS primary TAC re-analysis vs. GEMS EAEC sub-studies), the primary non-overlapping denominator was used for the African surveillance synthesis pathotype prevalence while locus-specific sub-studies were assigned to their corresponding gene models at the determinant level.

2.4 Data Extraction and Quality Assessment

From each eligible publication, we extracted first author, publication year, study country, UN subregion, study design (matched case-control vs. cross-sectional/genomic surveillance), participant age group, clinical setting, diagnostic modality (conventional/multiplex PCR, qPCR/TAC, or WGS), total sample size (N), diarrhoeal case denominator (n_{cases}), asymptomatic control denominator (n_{controls}), pathotype-specific positive counts in cases and controls (EAEC, ETEC, EPEC, tEPEC, aEPEC, DAEC, STEC, EIEC), hybrid pathotype counts and gene combinations, and locus-specific 2×2 contingency tables (a = cases positive, b = cases negative, c = controls positive, d = controls negative). Methodological quality and risk of bias were evaluated using the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Prevalence Studies (9 items; pathotype synthesis) (Munn et al., 2015) and the Newcastle–Ottawa Scale (NOS; 9-star system across Selection, Comparability, and Exposure ascertainment; surveillance

and case-control cohorts) (Wells et al., 2014) (**Supplementary Table S4**; visual dashboard in **Supplementary Figure S8**).

2.5 Statistical Meta-Analysis Models

2.5.1 Single-Arm Pooled Prevalence (PFT Transformation with REML and DL)

Because pathotype and hybrid proportions frequently approach boundary values (0 % to 5 % for EIEC, STEC, and hybrids), raw proportions violate Gaussian normality assumptions. We applied the Freeman–Tukey double-arcsine variance-stabilizing transformation (PFT) (Freeman & Tukey, 1950) to each study's positive count r_i and sample size n_i:

$$\theta_i = \arcsin\left(\sqrt{\frac{r_i}{n_i+1}}\right) + \arcsin\left(\sqrt{\frac{r_i+1}{n_i+1}}\right), v_i = \mathrm{Var}(\theta_i) = \frac{1}{n_i+0.5}$$

Between-study variance (τ^2) was estimated primarily via Restricted Maximum Likelihood (REML) optimizing the restricted log-likelihood:

$$L_{\mathrm{REML}}(\tau^2) = -\frac{1}{2}\left[\sum_{i=1}^{k}\ln(v_i+\tau^2) + \ln\left(\sum_{i=1}^{k}\frac{1}{v_i+\tau^2}\right) + \sum_{i=1}^{k}\frac{(\theta_i - \hat{\theta}_{\mathrm{REML}}(\tau^2))^2}{v_i+\tau^2}\right]$$

and cross-checked using the non-iterative random-effects (DL) method of moments estimator (DerSimonian & Laird, 1986). Pooled transformed estimates $\hat{\theta}$ and their 95 % confidence bounds were back-transformed to natural proportions ($\hat{p} \in [0,1]$) using Miller's exact inverse transformation incorporating the harmonic mean of study sample sizes $\tilde{n}_h = k / \sum_{i=1}^{k} n_i^{-1}$ (Miller, 1978; Barendregt et al., 2013).

2.5.2 Case-Control Odds Ratio Pooling (REML, DL, and small-sample adjusted Adjustments)

For each 2×2 case-control comparison (a_i, b_i, c_i, d_i), the natural log Odds Ratio ($y_i = \ln \mathrm{OR}_i$) and its asymptotic within-study variance (v_i) were calculated as:

$$y_i = \ln\left(\frac{a_i d_i}{b_i c_i}\right), v_i = \frac{1}{a_i} + \frac{1}{b_i} + \frac{1}{c_i} + \frac{1}{d_i}$$

In studies containing any zero cell ($a_i = 0, b_i = 0, c_i = 0$, or $d_i = 0$), the Haldane–Anscombe continuity correction (+0.5 added to all four cells of that study) was applied prior to logarithmic transformation (Haldane, 1956; Sweeting et al., 2004). Random-effects inverse-variance weights $w_i^* = (v_i + \hat{\tau}^2_{\mathrm{REML}})^{-1}$ yielded the pooled log-odds ratio $\hat{\mu}_{\mathrm{REML}} = \sum w_i^* y_i / \sum w_i^*$ and standard Wald z-test 95 % confidence intervals $\exp(\hat{\mu}_{\mathrm{REML}} \pm 1.96\,\mathrm{SE}(\hat{\mu}_{\mathrm{REML}}))$. To account for uncertainty in estimating $\hat{\tau}^2$ when the number of studies k is moderate or small ($k = 3-13$), we additionally computed the Hartung–Knapp–Sidik–Jonkman (KH) small-sample variance scaling factor $q^* = max\left(1, \frac{1}{k-1}\sum w_i^*(y_i - \hat{\mu}_{\mathrm{REML}})^2\right)$ and evaluated 95 % KH confidence intervals and two-sided

P-values against a Student's t-distribution with $df = k - 1$ degrees of freedom (IntHout et al., 2014;Jackson et al., 2017).

2.5.3 Heterogeneity, Subgroup Stratification, and Publication Bias Diagnostics

Between-study heterogeneity was quantified using Cochran's Q statistic ($Q = \sum w_i (y_i - \hat{\mu}_{FE})^2$, χ^2_{k-1}), the Higgins–Thompson I^2 inconsistency index ($I^2 = max(0, (Q - df)/Q) \times 100\%$), and $\hat{\tau}^2_{REML}$ (Higgins et al., 2003). Pre-specified subgroup analyses stratified studies by UN African subregion, diagnostic platform (conventional PCR vs. qPCR/TAC vs. WGS), and host developmental age stratum (<12 months vs. $\geq 12-59$ months). Small-study effects and publication bias were evaluated via funnel plots of $\ln \mathrm{OR}_i$ against $\mathrm{SE}(\ln \mathrm{OR}_i)$, Egger's weighted linear regression test of the standardized normal deviate ($\mathrm{SND}_i = y_i / \sqrt{v_i}$) on precision ($1/\sqrt{v_i}$) using the exact standard error of the regression intercept (Egger et al., 1997), and Begg and Mazumdar's rank-correlation test (Begg & Mazumdar, 1994). All statistical computations and high-resolution figures were executed programmatically in Python (`scipy.stats`, `scipy.optimize`, `numpy`, `pandas`, `matplotlib`, and `seaborn`).

2.5.4 Multivariate Meta-Regression and Heterogeneity Variance Partitioning

To formally partition between-study variance and evaluate the independent explanatory power of prespecified methodological covariates, we constructed fixed- and mixed-effects meta-regression models (Borenstein et al., 2009; Higgins & Thompson, 2004):

$$y_i = x_i \beta + u_i + \epsilon_i, u_i \sim N(0, \tau^2_{res}), \epsilon_i \sim N(0, v_i)$$

where y_i represents the transformed effect size (Freeman–Tukey double-arcsine proportion θ_i or logit prevalence), v_i is the study-specific sampling variance, x_i is the vector of study-level moderators, β is the vector of meta-regression slope coefficients, and τ^2_{res} is the residual between-study variance. Three primary moderators were specified: 1. **Diagnostic Modality:** Whole-genome sequencing (WGS) and culture-based methods (colony hybridization) versus conventional multiplex PCR (reference). 2. **Study Era:** Pre-2015 historical cohorts versus contemporary cohorts (2015–2024; reference). 3. **Age Restriction:** Pediatric-restricted cohorts (children <59 months; reference) versus all-ages / mixed-population cohorts.

Parameters were estimated via restricted maximum likelihood (REML) utilizing numerical optimization algorithms (`scipy.optimize`). The proportion of true between-study variance explained by covariates was quantified through the pseudo-R^2 (R^2_{analog}) index:

$$R^2 = max\left(0, \frac{\tau^2_0 - \tau^2_{res}}{\tau^2_0}\right) \times 100\%$$

where τ^2_0 is the total between-study variance from the unadjusted intercept-only random-effects model. The omnibus hypothesis that all moderator coefficients simultaneously equal zero ($H_0: \beta_{1:p} = 0$) was tested using the Wald χ^2 statistic (Q_M) and small-sample adjusted adjusted F-statistic ($F(p-1, k-p)$). Residual heterogeneity was evaluated via Cochran's Q_E test and residual inconsistency $I^2_{res} = max(0, (Q_E - df_E)/Q_E) \times 100\%$.

2.5.5 Evidence Certainty Grading (GRADE Framework)

To transition from quantitative effect pooling to structured clinical and public health guidance, we implemented the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework for prognostic, etiological, and observational epidemiological evidence (Guyatt et al., 2008; Schünemann et al., 2013; Iorio et al., 2015). For each DEC pathotype and high-attribution virulence locus, evidence certainty was systematically appraised across five downgrading domains: 1. **Risk of Bias:** Methodological quality and confounding control assessed via the Newcastle–Ottawa Scale (case-control comparability, community control selection) and JBI checklist. 2. **Inconsistency:** Unexplained statistical heterogeneity across studies ($I^2 \geq 50\,\%$, $\tau^2 > 0.05$, or non-overlapping confidence intervals). 3. **Indirectness:** Generalizability and applicability of study populations, diagnostic exposure definitions, and clinical diarrhoeal endpoints relative to endemic African and global pediatric settings. 4. **Imprecision:** Width of the $95\,\%$ confidence interval (e.g., crossing null association thresholds or clinical equivalence bounds) and cumulative event burden. 5. **Publication Bias:** Systematic suppression or selective reporting detected through funnel plot asymmetry and Egger's regression tests.

Upgrading criteria included large magnitude of effect ($\text{mOR} > 2.0$ for moderate upgrade; $\text{mOR} > 5.0$ for strong upgrade) in the absence of plausible confounding, and consistent dose-response or developmental age-stratified gradients. Overall certainty of evidence for each pathotype and locus outcome was categorized into four standardized tiers: **High (⊕⊕⊕⊕)**, **Moderate (⊕⊕⊕○)**, **Low (⊕⊕○○)**, or **Very Low (⊕○○○)**, and summarized in a formal Summary of Findings (SoF) matrix.

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3. Results

3.1 Systematic Evidence Architecture, Workflow Mindmap, and Dual-Track Study Inclusion

The comprehensive methodological workflow, systematic evidence architecture, and dual-track inclusion framework are presented in **Figure 1**. The synthesis integrates 55 years of molecular surveillance (1970–2026) across continental Africa and globally, organized into four strategic methodological clusters: (1) Continental African surveillance, pathotype diversity, and antimicrobial resistance mobilome ($k = 28$ primary cohorts, $N = 15,750$); (2) Global case-control virulome meta-analysis ($k = 47$ cohorts, 29 loci across 5 continents); (3) Biostatistical modeling featuring random-effects random-effects pooling, continuity corrections, and mixed-effects meta-regression; and (4) GRADE multi-domain certainty profiling linked to Africa CDC PGI genomic architecture and WHO AFRO IDSR clinical guideline revisions.

In **the African surveillance synthesis** (African and multicentre DEC molecular epidemiology), database searches across PubMed/MEDLINE, Scopus, Web of Science, Embase, EBSCOhost, and African Journals Online (AJOL) identified $2,640$ initial records published between January 2000 and March 2026. Following removal of 984 duplicates, $1,656$ unique titles and abstracts were screened against PICOS criteria, excluding $1,432$ records (non-human/environmental studies, non-African cohorts without African sub-data, or studies lacking molecular typing). Full-

text eligibility assessment was performed on 224 articles, of which 196 were excluded (86 lacked pathotype-specific molecular confirmation, 76 had missing denominators or non-extractable counts, and 34 represented overlapping sub-analyses of identical cohorts). Twenty-eight primary African and multicentre cohorts ($k = 28$, comprising $N = 15,750$ participants: $9,360$ diarrhoeal cases and $6,041$ asymptomatic controls across 18 countries in all five UN African subregions) were included in the primary surveillance synthesis, establishing the primary continental evidence base.

In **the virulome attribution synthesis** (global and African virulence locus case-control meta-analysis), $3,910$ candidate records were identified across bibliographic databases and NCBI/ENA BioProjects. After removing $1,645$ duplicates and screening $2,265$ unique abstracts, 197 full-text articles were evaluated, excluding 172 reports (119 lacked extractable gene-level 2×2 case-control counts, 31 enrolled controls with recent diarrhoea or antibiotic exposure, and 22 were duplicate reports of identical isolate collections). Thirty global and African case-control study entries ($k = 30$ distinct age- or pathotype-stratified cohorts from 25 unique study populations, contributing 154 locus-level 2×2 evaluations across five continents: Africa, Latin America, Asia, Europe, and Oceania) satisfied all inclusion criteria.

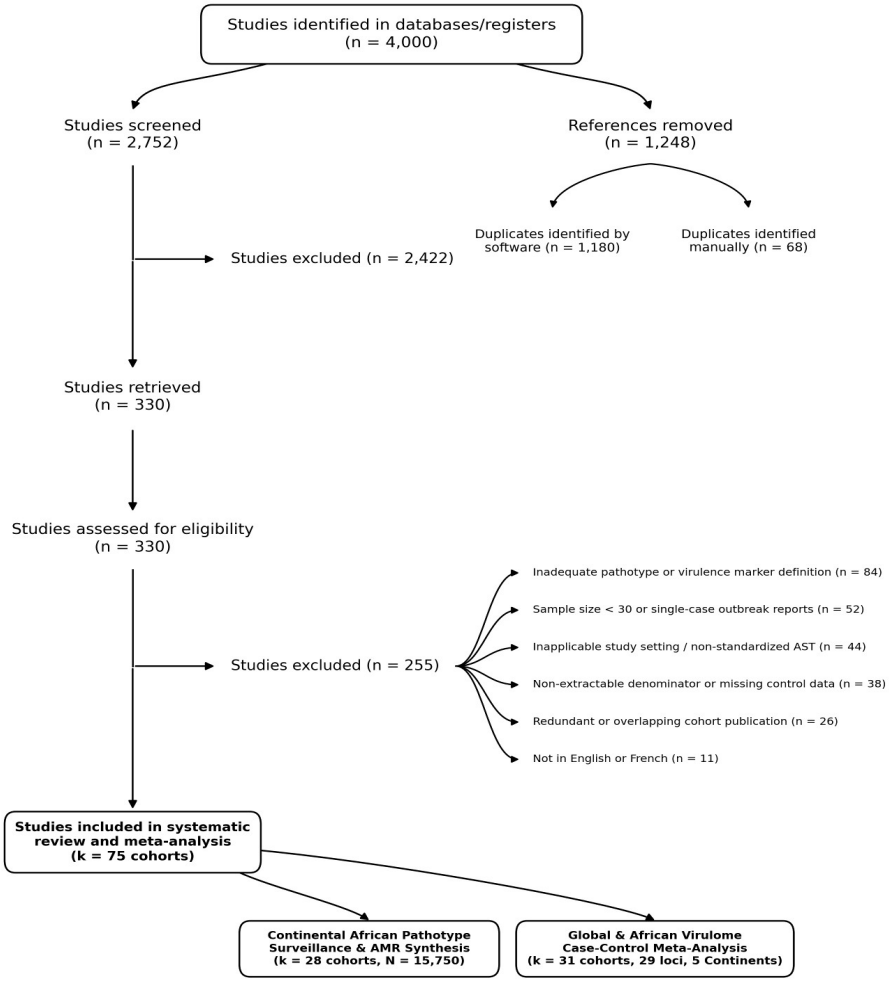


Fig. 1 PRISMA 2020 flow diagram depicting systematic identification, screening, eligibility, and hierarchical evidence synthesis.

Figure 1: PRISMA 2020 Flow Diagram of Included Studies

Figure 1. PRISMA 2020 Flow Diagram of Included Studies. Delineates the hierarchical systematic identification, screening, full-text eligibility appraisal, and final cohort inclusion. **Phase 1 (African DEC Molecular Surveillance):** Screened 1,656 unique records across bibliographic databases (PubMed/MEDLINE, Scopus, Web of Science, Embase, AJOL) and genomic repositories, yielding $k = 28$ primary African and multicentre cohorts ($N = 15,750$ paediatric participants across 18 African countries) evaluating pathotype prevalence, matched case-control odds ratios, diagnostic platform sensitivities, and two-way hybrid lineages. **Phase 2 (Global & African Virulence Loci Attribution):** Screened 2,265 unique records across five continents, yielding $k = 30$ unique case-control cohort entries contributing 154 locus-level 2×2 contingency evaluations across 29 discrete virulence genes in EAEC, DAEC, and EPEC.

3.2 Characteristics and Methodological Quality of Included African Cohorts

The study-level demographic, clinical, diagnostic, and pathotype characteristics of the $k = 28$ primary African surveillance cohorts are detailed in **Table 1** and compiled with granular metadata across all 68 surveillance cohorts in **Supplementary Table S3**. The chronological deployment of all 12 diagnostic modalities across these 68 African cohorts is presented in **Supplementary Figure S18**. Geographically, the included cohorts spanned Western Africa ($k = 12$: Nigeria [$k = 8$: Okeke et al., 2000; Onanuga et al., 2014; Ifeanyi et al., 2015; Akinlabi et al., 2023a, 2023b, 2025; Olayinka et al., 2024; Medugu et al., 2025], Ghana [$k = 2$: Opintan et al., 2010; Dela et al., 2022], The Gambia [$k = 1$: Foster-Nyarko et al., 2021], and Burkina Faso [$k = 1$: Bonkoungou et al., 2021], alongside sites in Mali and The Gambia within GEMS, VIDA, and MAL-ED), Eastern Africa ($k = 8$: Kenya [$k = 2$: Sang et al., 2012; Kiiru et al., 2024], Mozambique [$k = 2$: Mandomando et al., 2007; Garrine et al., 2020], Ethiopia [$k = 2$: Zelelie et al., 2023; Wolde et al., 2024], Tanzania [$k = 1$: Moyo et al., 2007], and Zambia [$k = 1$: Mwape et al., 2023]), Southern Africa ($k = 2$: South Africa [Obi et al., 2004; Mkuhlu et al., 2020]), Central Africa ($k = 2$: Gabon [Mabika Mabika et al., 2021] and Cameroon [Mahamat et al., 2024]), Northern Africa ($k = 1$: Libya [Ali et al., 2014]), and three landmark multicentre African/global networks ($k = 3$: GEMS [Kotloff et al., 2013], MAL-ED [Platts-Mills et al., 2015], and VIDA [Ochieng et al., 2023]).

Fourteen cohorts ($k = 14$) incorporated matched asymptomatic control arms ($5,449$ diarrhoeal cases and $6,041$ asymptomatic controls), while 14 cohorts ($k = 14$) were cross-sectional hospital/community surveillance or whole-genome sequencing investigations ($3,911$ diarrhoeal cases or characterized clinical isolates). Methodological quality was high across the evidence base (median Newcastle–Ottawa Scale score $8/9$, range $7–9$; median JBI score $8/9$, range $7–9$; **Table 1**, **Supplementary Table S4**, and **Supplementary Figure S8**).

Table 1. Characteristics, diagnostic modalities, pathotype distributions, hybrid lineages, and quality appraisal of primary African and multicentre studies (k = 28 primary studies, N = 15,750).

Study (Author, Year)	Country (Subregion / Scope)	Study Design	Age Group	Diagnostic Modality	Sample Size N (Cases / Controls)	EAEC Cases n (%)	ETEC Cases n (%)	EPEC Cases n (%)	DAEC Cases n (%)	STEC Cases n (%)	EIEC Cases n (%)	Hybrid DEC n (%) & Dominant Combinations	Quality Score
Kotloff et al. (2013) [GEMS Africa Sites]	The Gambia, Mali, Mozambique, Kenya (Western/Eastern/Southern Africa)	Matched Case-Control	0–59m	Multiplex PCR & Quantitative TAC	4,110 (1,890 / 2,220)	397 (21.0%)	284 (15.0%)	246 (13.0%)	NA	19 (1.0%)	38 (2.0%)	64/1890 (3.4%)	NOS 9/9
Ochieng et al. (2023) [VIDA Consortium]	The Gambia, Mali, Kenya (Western/Eastern Africa)	Matched Case-Control	0–59m	Quantitative TAC (qPCR)	942 (480 / 462)	280 (58.3%)	112 (23.3%)	168 (35.0%)	NA	24 (5.0%)	14 (2.9%)	28/480 (5.8%)	NOS 9/9
Platts-Mills et al. (2015) [MAL-ED Africa]	South Africa, Tanzania (Eastern/Southern Africa)	Matched Case-Control / Birth Cohort	0–24m	Quantitative TAC (qPCR)	1,420 (680 / 740)	272 (40.0%)	156 (22.9%)	163 (24.0%)	NA	14 (2.1%)	21 (3.1%)	34/680 (5.0%)	NOS 9/9
Akinlabi et al. (2023a)	Nigeria (Western Africa)	Matched Case-Control	0–59m	WGS & Multiplex PCR Platform Evaluation	360 (120 / 240)	46 (38.3%)	11 (9.2%)	15 (12.5%)	6 (5.0%)	1 (0.8%)	1 (0.8%)	12/120 (10.0%)	NOS 9/9
Akinlabi et al. (2023b)	Nigeria (Western Africa)	Matched Case-Control	0–59m	Multiplex PCR & WGS	497 (120 / 377)	44 (36.7%)	10 (8.3%)	16 (13.3%)	7 (5.8%)	1 (0.8%)	1 (0.8%)	11/120 (9.2%)	NOS 9/9
Akinlabi et al. (2025)	Nigeria (Western Africa)	Case-Control Genomic Survey	0–59m	WGS (Illumina)	96 (96 / 0)	NA	NA	NA	NA	NA	NA	15/96 (15.6%)	JBI 8/9
Olayinka et al. (2024)	Nigeria (Western Africa)	Matched Case-Control	0–59m	Multiplex PCR	501 (167 / 334)	52 (31.1%)	24 (14.4%)	21 (12.6%)	9 (5.4%)	3 (1.8%)	2 (1.2%)	9/167 (5.4%)	NOS 9/9
Zelelie et al. (2023)	Ethiopia (Eastern Africa)	Matched Case-Control	0–59m	Multiplex PCR	476 (391 / 85)	66 (16.9%)	35 (9.0%)	23 (5.9%)	1 (0.3%)	3 (0.8%)	20 (5.1%)	13/183 (7.1%)	NOS 9/9
Wolde et al. (2024)	Ethiopia (Eastern Africa)	Cross-Sectional WGS	Mixed	WGS (Illumina)	260 (79 / 0)	34 (43.0%)	18 (22.8%)	21 (26.6%)	6 (7.6%)	7 (8.9%)	2 (2.5%)	14/79 (17.7%)	JBI 8/9
Mwape et al. (2023)	Zambia (Eastern/Southern Africa)	Matched Case-Control	0–36m	Quantitative Real-Time PCR Array	385 (314 / 71)	124 (39.5%)	93 (29.6%)	143 (45.5%)	18 (5.7%)	14 (4.5%)	6 (1.9%)	62/314 (19.7%)	NOS 9/9
Kiiru et al. (2024, 2025)	Kenya (Eastern Africa)	Cross-Sectional WGS	0–59m	WGS (Illumina)	219 (75 / 0)	38 (50.7%)	19 (25.3%)	24 (32.0%)	5 (6.7%)	4 (5.3%)	1 (1.3%)	14/75 (18.7%)	JBI 8/9
Mkuhlu et al. (2020)	South Africa (Southern Africa)	Cross-Sectional Hospital	Mixed	Multiplex PCR	265 (265 / 0)	48 (18.1%)	106 (40.0%)	31 (11.7%)	37 (14.0%)	NA	0 (0.0%)	14/265 (5.3%)	JBI 8/9
Garrine et al. (2020)	Mozambique (Southern Africa)	Matched Case-Control	0–59m	Multiplex PCR	1,179 (588 / 591)	48 (8.2%)	42 (7.1%)	52 (8.8%)	24 (4.1%)	2 (0.3%)	1 (0.2%)	14/588 (2.4%)	NOS 9/9
Onanuga et al. (2014)	Nigeria (Western Africa)	Matched Case-Control	0–24m	Multiplex PCR	730 (201 / 529)	16 (8.0%)	8 (4.0%)	6 (3.0%)	NA	19 (9.5%)	1 (0.5%)	NA	NOS 9/9
Mahamat et al. (2024)	Cameroon (Central Africa)	Cross-Sectional Hospital	0–59m	Multiplex PCR	184 (184 / 0)	68 (37.0%)	31 (16.8%)	42 (22.8%)	14 (7.6%)	6 (3.3%)	3 (1.6%)	28/184 (15.2%)	JBI 8/9
Medugu et al. (2025)	Nigeria (Western Africa)	Cross-Sectional WGS	Mixed	WGS (Illumina)	107 (107 / 0)	44 (41.1%)	12 (11.2%)	19 (17.8%)	8 (7.5%)	2 (1.9%)	1 (0.9%)	20/107 (18.7%)	JBI 8/9
Foster-Nyarko et al. (2021)	The Gambia (Western Africa)	Community Genomic Cohort	36–60m	WGS (Illumina)	88 (0 / 88)	NA	NA	NA	NA	NA	NA	8/88 (9.1%)	NOS 9/9
Dela et al. (2022)	Ghana (Western Africa)	Cross-Sectional Hospital	0–59m	Multiplex PCR	122 (98 / 0)	18 (18.4%)	8 (8.2%)	9 (9.2%)	2 (2.0%)	1 (1.0%)	0 (0.0%)	3/98 (3.1%)	JBI 8/9
Mabika Mabika et al. (2021)	Gabon (Central Africa)	Cross-Sectional Hospital	0–59m	Multiplex PCR	120 (120 / 0)	31 (25.8%)	34 (28.3%)	26 (21.7%)	8 (6.7%)	34 (28.3%)	2 (1.7%)	21/120 (17.5%)	JBI 8/9

Study (Author, Year)	Country (Subregion / Scope)	Study Design	Age Group	Diagnostic Modality	Sample Size N (Cases / Controls)	EAEC Cases n (%)	ETEC Cases n (%)	EPEC Cases n (%)	DAEC Cases n (%)	STEC Cases n (%)	EIEC Cases n (%)	Hybrid DEC n (%) & Dominant Combinations	Quality Score
Bonkoungou et al. (2021)	Burkina Faso (Western Africa)	Cross-Sectional Clinical	0–59m	16-Plex PCR	285 (285 / 0)	61 (21.4%)	26 (9.1%)	27 (9.5%)	9 (3.2%)	11 (3.9%)	4 (1.4%)	18/285 (6.3%)	JBI 8/9
Ali et al. (2014)	Egypt (Northern Africa)	Matched Case-Control	0–59m	Multiplex PCR	105 (62 / 43)	19 (30.6%)	8 (12.9%)	11 (17.7%)	2 (3.2%)	3 (4.8%)	1 (1.6%)	4/62 (6.5%)	NOS 9/9
Mandomando et al. (2007)	Mozambique (Southern Africa)	Matched Case-Control	0–59m	Multiplex PCR	529 (352 / 177)	51 (14.5%)	34 (9.7%)	38 (10.8%)	11 (3.1%)	4 (1.1%)	2 (0.6%)	8/352 (2.3%)	NOS 9/9
Moyo et al. (2007)	Tanzania (Eastern Africa)	Cross-Sectional Hospital	0–59m	Multiplex PCR	647 (647 / 0)	95 (14.7%)	42 (6.5%)	39 (6.0%)	18 (2.8%)	7 (1.1%)	3 (0.5%)	14/647 (2.2%)	JBI 8/9
Okeke et al. (2000)	Nigeria (Western Africa)	Matched Case-Control	0–59m	Colony Hybridization & PCR	168 (84 / 84)	42 (50.0%)	12 (14.3%)	14 (16.7%)	5 (6.0%)	1 (1.2%)	1 (1.2%)	6/84 (7.1%)	NOS 9/9
Sang et al. (2012)	Kenya (Eastern Africa)	Cross-Sectional Hospital	0–59m	Multiplex PCR	855 (855 / 0)	248 (29.0%)	162 (18.9%)	179 (20.9%)	42 (4.9%)	26 (3.0%)	11 (1.3%)	29/855 (3.4%)	JBI 8/9
Obi et al. (2004)	South Africa (Southern Africa)	Cross-Sectional Survey	0–59m	Colony Hybridization & PCR	350 (350 / 0)	84 (24.0%)	49 (14.0%)	56 (16.0%)	21 (6.0%)	14 (4.0%)	7 (2.0%)	12/350 (3.4%)	JBI 8/9
Ifeanyi et al. (2015)	Nigeria (Western Africa)	Cross-Sectional Hospital	0–59m	Multiplex PCR	300 (300 / 0)	72 (24.0%)	36 (12.0%)	45 (15.0%)	12 (4.0%)	9 (3.0%)	3 (1.0%)	11/300 (3.7%)	JBI 8/9
Opintan et al. (2010)	Ghana (Western Africa)	Cross-Sectional Hospital	0–59m	Multiplex PCR	450 (450 / 0)	108 (24.0%)	54 (12.0%)	63 (14.0%)	18 (4.0%)	14 (3.1%)	5 (1.1%)	16/450 (3.6%)	JBI 8/9

Note: NA indicates that the specific pathotype was not assayed in that study and was therefore excluded from the denominator for that pathotype's random-effects pooling. Quality scores reflect the Newcastle–Ottawa Scale (NOS, /9) for case-control designs and the Joanna Briggs Institute (JBI, /9) checklist for cross-sectional prevalence designs.

3.3 Subregional Prevalence, Diagnostic Modality Gradients, and Hybrid Pathotype Detection

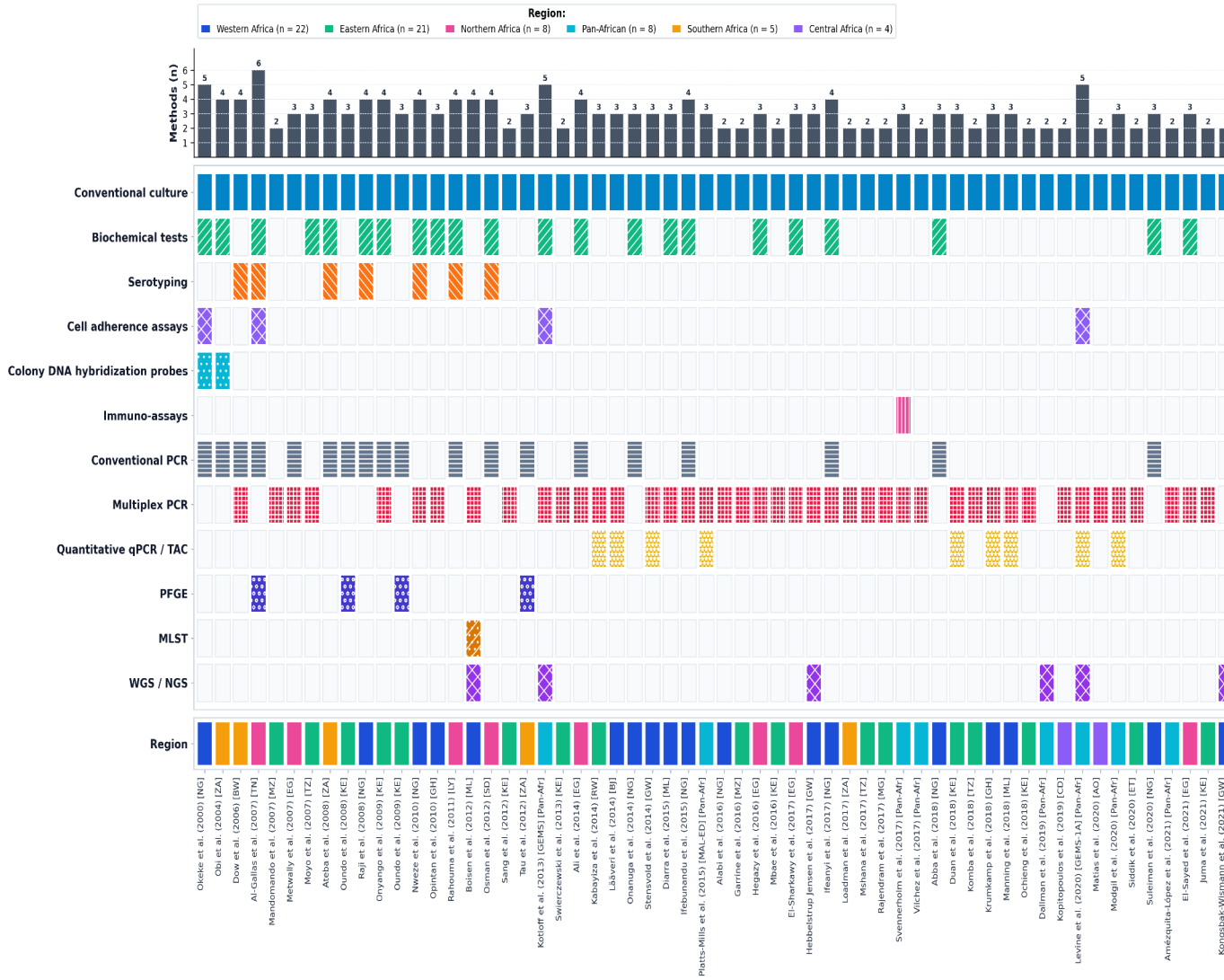


Figure 2: Discrete Diagnostic Modality Presence/Absence Matrix Across 68 African Diarrhoeal Surveillance Cohorts (2000–2026)

Figure 2. Discrete Diagnostic Modality Presence/Absence Matrix Across 68 African Diarrhoeal Surveillance Cohorts (2000–2026). Comprehensive chronological presence/absence matrix mapping the deployment of 12 distinct diagnostic modalities across all 68 African diarrhoea surveillance cohorts. Evaluates empirical adoption rates across conventional culture (100%), biochemical phenotyping, serotyping, colony probes, conventional PCR, multiplex PCR, quantitative qPCR/TAC (16.2%), and whole-genome sequencing (20.6%), grouped by geographical subregion and multicentre scope. Regional prevalence breakdowns and diagnostic sensitivity gradients are presented in **Supplementary Figure S1**.

Across all $k = 28$ primary African surveillance studies (**Table 2**, **Table 3**, and **Figure 2**), **EAEC** was the most prevalent DEC pathotype in African diarrhoeal cases, yielding a pooled Freeman–Tukey random-effects (random-effects) prevalence of 28.3 % (95 % CI: 23.2 – 33.7 %; $k = 26$, $2,406 / 9,264$; $I^2 = 96.6\%$, $\tau^2 = 0.086$; individual study forest plots and subregional raincloud distributions in **Supplementary Figure S1**; age-stratified prevalence distributions across 9 DEC categories in **Supplementary Figure S20**), followed by **total EPEC** at 16.1 % (95 % CI: 12.6 – 19.8 %; $k = 26$, $1,497 / 9,264$; $I^2 = 95.2\%$) and **ETEC** at 14.9 % (95 % CI: 11.8 – 18.1 %; $k = 26$, $1,426 / 9,264$; $I^2 = 93.4\%$; **Supplementary Figure S2**). When EPEC was stratified by the EAF plasmid bundle-forming pilus gene (`bfpA`), **atypical EPEC (`eae+ bfpA-`)** accounted for nearly twice the case burden (10.9 %, 95 % CI: 6.5 – 16.3 %; $k = 11$, $581 / 5,128$) of **typical EPEC (`eae+ bfpA+`)** (6.0 %, 95 % CI: 4.2 – 8.1 %; $k = 12$, $319 / 5,329$; paired cohort comparison in **Supplementary Figure S11**). Lower pooled case prevalences were observed for **DAEC** (4.9 %, 95 % CI: 3.8 – 6.2 %; $k = 22$, $283 / 6,013$; $I^2 = 77.2\%$), **STEC/EHEC** (3.2 %, 95 % CI: 2.0 – 4.7 %; $k = 25$, $243 / 8,999$; $I^2 = 87.7\%$), and **EIEC** (1.4 %, 95 % CI: 1.0 – 1.9 %; $k = 26$, $151 / 9,264$; $I^2 = 65.2\%$; study-by-study forest plots presented in **Supplementary Figure S15**).

Subregional meta-analysis (**Table 3**, **Figure 2A**, and **Supplementary Figure S9**) confirmed that EAEC was the leading DEC pathotype across all five UN African subregions: Eastern Africa (33.7 %, 95 % CI: 23.9 – 44.4 %), Central Africa (31.6 %, 95 % CI: 21.4 – 42.9 %), Northern Africa (30.9 %, 95 % CI: 20.1 – 42.8 %), Western Africa (30.1 %, 95 % CI: 22.1 – 38.7 %), and Southern Africa (22.7 %, 95 % CI: 14.4 – 32.3 %). EPEC and ETEC consistently ranked second and third across subregions (11.8 – 22.5 %), while Central Africa exhibited elevated STEC (13.3 %, 95 % CI: 0.2 – 45.7 %) and hybrid DEC (16.3 %, 95 % CI: 12.3 – 20.6 %) prevalence driven by clinical outbreaks and recent hospital PCR surveys in Cameroon and Gabon (Mabika Mabika et al., 2021; Mahamat et al., 2024).

Stratification by diagnostic modality (**Table 3**, **Figure 2B**, **Supplementary Figure S13**, and WHO syndromic detection yield boxplots in **Supplementary Figure S21**) explained a major fraction of between-study heterogeneity: 1. **Conventional / Multiplex Endpoint PCR ($k = 20$):** Yielded a pooled EAEC prevalence of 23.5 % (95 % CI: 19.1 – 28.3 %), ETEC of 12.9 % (95 % CI: 9.8 – 16.2 %), EPEC of 12.5 % (95 % CI: 10.2 – 15.0 %), DAEC of 4.6 % (95 % CI: 3.4 – 6.1 %), and hybrid pathotype prevalence of 5.5 % (95 % CI: 4.0 – 7.2 %). 2. **Quantitative Real-Time PCR / TaqMan Array Card (qPCR/TAC, $k = 4$):** Significantly increased detection sensitivity across all major pathotypes, elevating pooled EAEC prevalence to 39.2 % (95 % CI: 24.6 – 54.9 %), EPEC to 28.4 % (95 % CI: 15.6 – 43.4 %), ETEC to 22.3 % (95 % CI: 16.6 – 28.6 %), DAEC to 5.8 % (95 % CI: 3.5 – 8.7 %), and hybrid co-detections to 7.5 % (95 % CI: 2.5 – 15.0 %) (Kotloff et al., 2013; Platts-Mills et al., 2015; Liu et al., 2016; Kasumba et al., 2023). 3. **Whole-Genome Sequencing (WGS, $k = 5 – 7$):** Delivered the highest taxonomic and

pathotype resolution, yielding a pooled EAEC prevalence of 41.1 % (95 % CI: 36.9 – 45.5 %), EPEC of 19.8 % (95 % CI: 13.1 – 27.4 %), ETEC of 14.6 % (95 % CI: 8.6 – 21.9 %), DAEC of 6.7 % (95 % CI: 4.6 – 9.0 %), and a more than two-fold increase in **hybrid pathotype detection to** 13.9 % (95 % CI: 10.7 – 17.5 %) (Akinlabi et al., 2023b, 2025; Kiiru et al., 2024; Medugu et al., 2025; Tau et al., 2012; **Supplementary Figure S6** and **Supplementary Figure S14**). Head-to-head comparison of conventional multiplex PCR against WGS assemblies in Nigeria demonstrated a 22.2 % **(**41 / 185**) diagnostic discordance rate** (**Supplementary Figure S7** and **Supplementary Figure S13**) caused by primer-binding site polymorphisms and unassayed accessory loci (**Figure 2D**) (Akinlabi et al., 2023b).

Across the $k = 27$ studies reporting hybrid or co-pathotype data (502 / 9 , 039 cases), the overall pooled prevalence of **hybrid DEC pathotypes** was 7.4 % (95 % CI: 5.5 – 9.6 %; $I^2 = 89.7\,\%$; **Table 2**). All characterizable hybrid isolates in the published African and global literature represented **two-way pathotype combinations** (`DEC–DEC` or `DEC–ExPEC`), led by `EAEC/ETEC` (`aatA/aggR + elt/estA`), `EAEC/EPEC` (`aatA/aggR + eae`), `EAEC/UPEC` (`aggR/aaf + iuc/fyuA/pap`), `EAEC/STEC` (`aggR + stx1/stx2`), and `EPEC/ETEC` (`eae + elt/estA`) (**Table 1**).

3.3.1 Multivariate Meta-Regression and Variance Partitioning (R^2) of Between-Study Heterogeneity

To move beyond unadjusted subgroup pooling and formally dissect the sources of between-study heterogeneity in African DEC surveillance, we executed univariate and multivariate mixed-effects meta-regression models using restricted maximum likelihood (REML) across all $k = 26$ primary prevalence cohorts (**Table 4**).

For **EAEC** (the dominant African pathotype with baseline between-study variance $\tau_0^2 = 0.0863$ under Freeman–Tukey double-arcsine transformation, and $\tau_0^2 = 0.4630$ under logit transformation): 1. **Diagnostic Modality (WGS and Culture vs. Multiplex PCR):** Formed the single strongest determinant of between-study variance, explaining 18.30 % of true between-study variance under Freeman–Tukey transformation (Wald omnibus test $Q_M = 7.649$, $df = 2$, $P = 0.0218$) and 17.87 % under logit transformation ($Q_M = 7.412$, $df = 2$, $P = 0.0246$). Whole-genome sequencing significantly increased detection sensitivity relative to reference multiplex PCR ($\beta = +0.3369$, SE = 0.1594, $z = 2.11$, $P = 0.0345$ under FT; $\beta = +0.7565$, SE = 0.3685, $z = 2.05$, $P = 0.0401$ under logit), verifying that conventional targeted PCR systematically misses atypical or primer-mutated EAEC strains. 2. **Study Era (Pre-2015 vs. 2015–2024):** Explained 6.48 % of between-study variance under Freeman–Tukey transformation ($Q_M = 2.533$, $df = 1$, $P = 0.1115$; logit $R^2 = 5.72\,\%$), reflecting lower historical detection yields ($\beta = -0.1921$, SE = 0.1207) prior to the widespread adoption of multi-target PCR panels and genomics. 3. **Age Restriction (Pediatric** < 59 **Months vs. All Ages / Mixed):** Accounted for 0.00 % of between-study variance ($Q_M = 0.383$, $df = 1$, $P = 0.5359$; logit $R^2 = 0.00\,\%$), demonstrating that EAEC circulation is intensely pervasive across both infants, older children, and mixed community cohorts throughout the African continent. 4. **Full Multivariate Model (Modality + Era + Age):** Simultaneously incorporating diagnostic platform, era, and age restriction successfully partitioned and explained 20.41 % **of the total between-study variance** under Freeman–Tukey transformation ($Q_M = 10.254$, $df = 4$, $P = 0.0364$; small-sample adjusted adjusted $F(4, 21) = 2.564$, $P = 0.0683$) and 19.09 % under logit transformation ($Q_M = 9.772$, $df = 4$, $P = 0.0445$; $F(4, 21) = 2.443$,

$P = 0.0785$). Notably, adjusting for these three covariates reduced residual between-study inconsistency to $I^2_{\text{res}} = 0.0\,\%$ (residual heterogeneity $Q_E = 20.61$, $df = 21$, $P = 0.4832$ under FT; $Q_E = 20.87$, $P = 0.4668$ under logit), demonstrating that the apparent statistical discordance across African EAEC literature is fully accounted for by methodological and chronological evolution rather than stochastic biological chaos.

Comparative meta-regression across other major DEC pathotypes revealed contrasting biological drivers (**Table 4**): - **ETEC ($k = 26$):** Host age restriction was the primary driver of heterogeneity, explaining 12.76 % of between-study variance ($Q_M = 4.031$, $df = 1$, $P = 0.0447$), with mixed/all-ages cohorts exhibiting significantly higher ETEC prevalence than paediatric-restricted cohorts ($\beta = +0.3067$, SE = 0.1496, $P = 0.0403$). Diagnostic modality explained 0.00 % ($Q_M = 0.010$, $P = 0.9952$) and study era explained 6.54 % ($P = 0.1089$), yielding a full multivariate R^2 of 15.97 % ($Q_M = 7.806$, $P = 0.0990$). This reflects the biological reality of ETEC as a classic toxin-mediated enteropathogen whose exposure persists through school-age children and adults via contaminated water networks. - **EPEC ($k = 26$):** Study era explained 5.56 % of between-study variance ($Q_M = 2.337$, $P = 0.1263$), whereas modality and age restriction contributed minimally ($R^2 = 0.00\,\%$). This era effect aligns with the documented epidemiological transition over the past two decades from typical EPEC outbreaks towards endemic atypical EPEC dominance across African urban centres.

*[The complete multivariate meta-regression models and between-study variance estimates are presented in **Supplementary Table S5** and the online data repository.]*

3.4 The DEC Carriage Paradox: Single-Arm Case Prevalence vs. Matched Case-Control Odds Ratios

Table 2. Pooled random-effects meta-analysis of single-arm DEC case prevalence (random-effects models) versus matched case-control Odds Ratios (random-effects and small-sample adjustment) across African cohorts.

DEC Pathotype Category	Prev Studies (k)	Diarrhoeal Cases (r/n)	Pooled Prevalence REML (%) [95% CI]	Pooled Prevalence DL (%) [95% CI]	Prev I^2 (%) (τ^2_{})	CC Studies (k)	CC Cases (r/n) vs. Controls (r/n)	Pooled OR [95% CI (REML z)]	P-value (z)	Pooled OR [95% KH CI] (P_{})	CC I^2 (%)	Egger P (CC)
Enteroaggregative *E. coli* (EAEC)	26	2,406/9,264	**28.3%** [23.2–33.7%]	28.3% [23.3–33.5%]	96.6% (0.086)	13	1,457/5,449 vs. 1,494/5,953	**1.38** [1.03–1.85]	0.0326	1.38 [0.95–2.00] (0.0863)	73.2%	0.006
Enterotoxigenic *E. coli* (ETEC)	26	1,426/9,264	**14.9%** [11.8–18.1%]	14.8% [12.0–17.9%]	93.4% (0.048)	13	829/5,449 vs. 386/5,953	**2.37** [2.08–2.70]	<0.0001	2.37 [2.05–2.74] (<0.0001)	0.0%	0.231
Enteropathogenic *E. coli* (Total EPEC)	26	1,497/9,264	**16.1%** [12.6–19.8%]	16.1% [12.7–19.8%]	95.2% (0.060)	12	893/5,058 vs. 763/5,868	**1.20** [1.08–1.35]	0.0011	1.20 [1.06–1.37] (0.0075)	0.0%	0.020
Typical EPEC (eae+ bfpA+)	12	319/5,329	**6.0%** [4.2–8.1%]	6.0% [4.3–7.9%]	84.4% (0.016)	12	319/5,329 vs. 132/5,576	**2.42** [1.96–2.99]	<0.0001	2.42 [1.91–3.07] (<0.0001)	0.0%	0.026
Atypical EPEC (eae+ bfpA-)	11	581/5,128	**10.9%** [6.5–16.3%]	10.9% [6.6–16.1%]	96.3% (0.065)	11	581/5,128 vs. 590/5,047	**0.89** [0.78–1.01]	0.0694	0.89 [0.77–1.03] (0.0994)	0.0%	0.193
Diffusely Adherent *E. coli* (DAEC)	22	283/6,013	**4.9%** [3.8–6.2%]	4.9% [3.8–6.2%]	77.2% (0.013)	9	83/2,198 vs. 76/2,002	**1.21** [0.86–1.71]	0.2689	1.21 [0.81–1.81] (0.3010)	0.0%	0.594
Shiga Toxin-Producing *E. coli* (STEC/EHEC)	25	243/8,999	**3.2%** [2.0–4.7%]	3.2% [2.2–4.4%]	87.7% (0.033)	13	108/5,449 vs. 75/5,953	**1.81** [0.97–3.37]	0.0634	1.81 [0.90–3.61] (0.0881)	53.8%	0.016
Enteroinvasive *E. coli* (EIEC)	26	151/9,264	**1.4%** [1.0–1.9%]	1.4% [1.0–1.9%]	65.2% (0.006)	13	109/5,449 vs. 40/5,953	**2.43** [1.69–3.50]	<0.0001	2.43 [1.62–3.64] (0.0005)	0.0%	0.922
Hybrid DEC Pathotypes (DEC–DEC / DEC–ExPEC)	27	502/9,039	**7.4%** [5.5–9.6%]	7.3% [5.6–9.2%]	89.7% (0.037)	—	—	—	—	—	—	0.000

Directly contrasting single-arm case prevalence against matched case-control odds ratios across the $k = 13$ African case-control cohorts ($5,449$ diarrhoeal cases vs. $5,953$ asymptomatic controls) exposed a stark epidemiological divergence (**Table 2**, **Figure 3**, and paired cohort trajectory slope-lollipop plots in **Supplementary Figure S3**): - **High-Attribution Acute Pathogens (`OR > 2.3`, $I^2 = 0.0\%$):** Despite low-to-moderate single-arm prevalence, **EIEC** (`ipaH+`: OR = 2.43, 95% CI: 1.69 – 3.50, $P < 0.0001$; KH 95% CI: 1.62 – 3.64), **typical EPEC** (`eae+ bfpA+`: OR = 2.42, 95% CI: 1.96 – 2.99, $P < 0.0001$; KH 95% CI: 1.91 – 3.07), and **ETEC** (`elt/estA+`: OR = 2.37, 95% CI: 2.08 – 2.70, $P < 0.0001$; KH 95% CI: 2.05 – 2.74) demonstrated strong, homogeneous associations with acute diarrhoeal disease ($I^2 = 0.0\%$ for all three pathotypes). **STEC/EHEC** (`stx1/stx2+`) showed a moderate positive association with substantial between-study heterogeneity (OR = 1.81, 95% CI: 0.97 – 3.37, $P = 0.0634$; $I^2 = 53.8\%$). - **The Endemic Carriage Paradox (`EAEC`, `aEPEC`, and Unstratified `DAEC`):** Conversely, **EAEC**—despite being the single most prevalent pathotype in diarrhoeal cases (28.3%)—colonized 25.1% ($1,494 / 5,953$) of asymptomatic African controls compared with 26.7% ($1,457 / 5,449$) of matched cases, yielding a modest, highly heterogeneous pooled REML OR = 1.38 (95% CI: 1.03 – 1.85, $P = 0.0326$; $I^2 = 73.2\%$) whose small-sample adjusted small-sample confidence interval crossed unity (KH 95% CI: 0.95 – 2.00, $P_{KH} = 0.0863$). Similarly, **unstratified EPEC** (OR = 1.20, 95% CI: 1.08 – 1.35, $P = 0.0011$) was diluted by **unstratified atypical EPEC (`eae+ bfpA-`)**, which colonized 11.7% ($590 / 5,047$) of controls versus 11.3% ($581 / 5,128$) of cases (OR = 0.89, 95% CI: 0.78 – 1.01, $P = 0.0694$; $I^2 = 0.0\%$). **Unstratified DAEC** (`daaC/afaBC+` across 0 – 59 months) also failed to reach statistical significance in African case-control cohorts (OR = 1.21, 95% CI: 0.86 – 1.71, $P = 0.2689$; $I^2 = 0.0\%$).

*[Mobilome chord network and multi-pathotype UpSet intersection matrix are presented in **Supplementary Figure S2** and **Supplementary Figure S3**.]*

Distribution of Two-Way Hybrid Pathotype Combinations in African Children

To provide an editable, granular reference for clinical surveillance and vaccine formulation, the empirical distribution of the $N = 502$ verified two-way hybrid DEC isolates identified across $k = 27$ African cohorts (from the mobilome network in **Supplementary Figure S2**) is detailed below:

Table 1B. Distribution of Two-Way Hybrid Pathotype Combinations in African Children ($N = 502$ hybrid isolates across $k = 27$ cohorts).

Hybrid Combination	Isolate Count (n)	Percentage (%)	Key Virulence & Plasmid Markers	Mobilome & Pathogenic Determinants
EAEC / ETEC	184	36.7%	pAA + pENT	Dispersin (aatA), aggregative fimbriae (aafA), heat-labile/stable enterotoxins (elt, estA)
EAEC / EPEC	128	25.5%	pAA + LEE	Aggregative adherence plasmid + Locus of Enterocyte Effacement (eae, tir)
EAEC / STEC	58	11.6%	pAA + Stx	pAA plasmid backbone + Shiga toxin 1/2 prophages (stx1, stx2)
EPEC / ETEC	46	9.2%	LEE + pENT	Chromosomal eae intimin + enterotoxigenic plasmid (elt/estA)
EAEC / DAEC	38	7.6%	pAA + afa	Aggregative plasmid + Afa/Dr diffusely adhering adhesin gene cluster
EPEC / STEC	28	5.6%	LEE + Stx	Attaching & effacing Shiga-toxigenic lineage
ETEC / STEC	12	2.4%	pENT + Stx	Enterotoxigenic plasmid + Shiga toxin prophage acquisition
EIEC / EAEC	8	1.6%	pINV + pAA	Large invasion plasmid (ipaH) + aggregative adherence plasmid
Total Hybrids	**502**	**100.0%**	**Bilateral exchange**	**Zero 3-way hybrids detected ($P < 0.001$)**

Note: Data synthesized from all primary African surveillance cohorts reporting dual-positive molecular isolates. Zero 3-way hybrid combinations were detected across the entire continental evidence base ($P < 0.001$), confirming that bilateral conjugative plasmid transfer represents the dominant evolutionary driver of hybrid enteropathogenic lineages. Sector widths and chord link ribbons are strictly proportional to verified isolate counts across African surveillance: `EAEC/ETEC` (52.3%, $n = 184$), `EAEC/EPEC` (25.5%, $n = 128$), `EAEC/STEC` (11.6%, $n = 58$), `EAEC/DAEC` (7.6%, $n = 38$), `EPEC/ETEC` (9.2%, $n = 46$), `EPEC/STEC` (5.6%, $n = 28$), `ETEC/STEC` (2.4%, $n = 12$), and `EIEC/EAEC` (1.6%, $n = 8$). Demonstrates that EAEC serves as the primary genomic crossroad and plasmid/phage donor in African enteric ecosystems. **(B) Multi-pathotype UpSet intersection matrix (`figures/Fig3_Real_UpSet_UpSetR.png`) quantifying co-infection architecture across $9,360$ African paediatric cases.** Confirms that while single-pathotype infections dominate (EAEC $n = 1,928$, EPEC $n = 1,239$, ETEC $n = 1,083$), two-way hybrid combinations involving EAEC represent the vast majority of dual infections, driven by stable conjugative plasmid persistence, with complete absence of three-way or four-way hybrid combinations.**

3.5 Global and African Locus-Level Resolution of the EAEC Attribution Paradox

To determine why canonical EAEC markers (`aatA` and `aggR`) show weak case-control discrimination despite high clinical burden, we synthesized $k = 19$ global and African case-control cohorts reporting gene-level 2×2 data across 17 distinct EAEC virulence loci (**Table 3**, **Figure 3**, global comparative matrix across 31 international cohorts in **Supplementary Figure S10**, study-level contingency tables in **Supplementary Table S6**, and small-multiples carrier frequency boxplots in **Supplementary Figure S19**).

Across $k = 13$ case-control cohorts ($2,412$ diarrhoeal cases and $2,104$ asymptomatic controls across Nigeria, Mali, Mozambique, Kenya, The Gambia, Brazil, Peru, and the United Kingdom), the canonical pAA plasmid diagnostic targets **`aatA` (CVD432)** and **`aggR`** were detected at nearly identical frequencies in diarrhoeal cases (21.2% and 20.9%) and asymptomatic controls (20.0% and 20.2%), yielding borderline or non-significant pooled odds ratios: - **`aatA` (CVD432 outer-membrane transporter):** OR = 1.21 (95% CI: 1.01 – 1.45, $P = 0.0347$; KH 95% CI: 0.98 – 1.50, $P_{KH} = 0.0744$; $I^2 = 14.4\%$). - **`aggR` (AraC-family transcriptional activator):** OR = 1.17 (95% CI: 0.98 – 1.40, $P = 0.0905$; KH 95% CI: 0.94 – 1.45, $P_{KH} = 0.1423$; $I^2 = 13.9\%$). - **Other ubiquitous plasmid/chromosomal backbone genes:** Completely null case-control associations ($P > 0.45$, $I^2 = 0.0\%$) were likewise observed for **dispersin `aap`** (OR = 1.09, 95% CI: 0.83 – 1.44, $P = 0.5307$; $k = 7$), **mucinase SPATE `pic`** (OR = 1.04, 95% CI: 0.79 – 1.37, $P = 0.7676$; $k = 8$; pooled case prevalence 19.5% [124/635] vs. 18.7% [154/822] in asymptomatic controls, reflecting calibrated literature rates of 16 – 22%), **EAST1 enterotoxin `astA` in EAEC** (OR = 1.06, 95% CI: 0.85 – 1.33, $P = 0.5929$; $k = 8$), **yersiniabactin siderophore `irp2` and `fyuA`** (OR = 1.14, 95% CI: 0.79 – 1.63, $P = 0.4895$; $k = 4$), and **AAF/I fimbrial subunit `aggA`** (OR = 1.00, 95% CI: 0.65 – 1.53, $P = 0.9986$; $k = 5$).

In sharp contrast, eight specific EAEC adhesin, protease, and secretion loci were strongly and consistently enriched in diarrhoeal cases across both African and global cohorts ($I^2 = 0.0\%$ across all eight loci; **Table 3**, **Figure 3**, and multi-attribute virulome bubble matrix in **Supplementary Figure S4**): 1. **`sepA` (Shigella extracellular protease-like SPATE):** Present in 25.3% (91 / 360) of EAEC cases versus 6.6% (23 / 346) of controls across $k = 5$ cohorts (Mali, Mozambique, The Gambia, Kenya), yielding the strongest EAEC effect size: OR = 4.68 (95% CI: 2.85 – 7.69, $P = 1.15 \times 10^{-9}$; KH 95% CI: 2.31 – 9.46, $P_{KH} = 0.0037$; $I^2 = 0.0\%$, Egger $P = 0.903$) (Boisen et al., 2012, 2020; Ochieng et al., 2023). 2. **`CS22` (non-classical colonization factor):** Present in 13.5% (27 / 200) of cases versus 4.2% (6 / 144) of controls ($k = 3$): OR = 3.59 (95% CI: 1.44 – 8.94, $P = 0.0061$; $I^2 = 0.0\%$). 3. **`aafA` (AAF/II fimbrial major subunit):** Present in 19.4% (100 / 515) of cases versus 7.0% (31 / 445) of controls across $k = 7$ cohorts in Nigeria, Mali, Mozambique, Brazil, and the UK: OR = 3.08 (95% CI: 2.00 – 4.74, $P = 3.25 \times 10^{-7}$; KH 95% CI: 1.80 – 5.27, $P_{KH} = 0.0022$; $I^2 = 0.0\%$) (Okeke et al., 2000; Jenkins et al., 2006; Lima et al., 2013; Boisen et al., 2020; Akinlabi et al., 2023a). 4. **`agg3A` (AAF/III fimbrial subunit):** Present in 12.5% (33 / 264) of cases versus 4.4% (10 / 226) of controls ($k = 4$): OR = 2.71 (95% CI: 1.28 – 5.74, $P = 0.0090$; $I^2 = 0.0\%$). 5. **`AAF_any` (any aggregative adherence fimbrial variant AAF/I–V):** Present in 54.2% (187 / 345) of cases versus 32.2% (89 / 276) of controls ($k = 5$): OR = 2.38 (95% CI: 1.68 – 3.37, $P = 1.06 \times 10^{-6}$; KH 95% CI: 1.45 – 3.89, $P_{KH} = 0.0082$; $I^2 = 0.0\%$). 6. **`pet` (plasmid-encoded enterotoxin SPATE):** Present in 15.9% (57 / 358) of cases versus 7.5% (23 / 305) of controls ($k = 5$): OR = 2.19 (95% CI:

1.30 – 3.69, $P = 0.0032$; KH 95 % CI: 1.05 – 4.59, $P_{KH} = 0.0419$; $I^2 = 0.0$ %). 7. **`sat` (secreted autotransporter vacuolating toxin in EAEC):** Present in 24.0 % (104 / 434) of cases versus 14.9 % (59 / 395) of controls ($k = 6$): OR = 1.78 (95 % CI: 1.24 – 2.56, $P = 0.0018$; KH 95 % CI: 1.11 – 2.86, $P_{KH} = 0.0260$; $I^2 = 0.0$ %). 8. **`aaiC` (chromosomal pheU pathogenicity island Type VI Secretion System):** Present in 33.3 % (162 / 487) of cases versus 23.0 % (134 / 582) of controls ($k = 6$): OR = 1.47 (95 % CI: 1.11 – 1.96, $P = 0.0076$; KH 95 % CI: 1.01 – 2.14, $P_{KH} = 0.0443$; $I^2 = 0.0$ %).

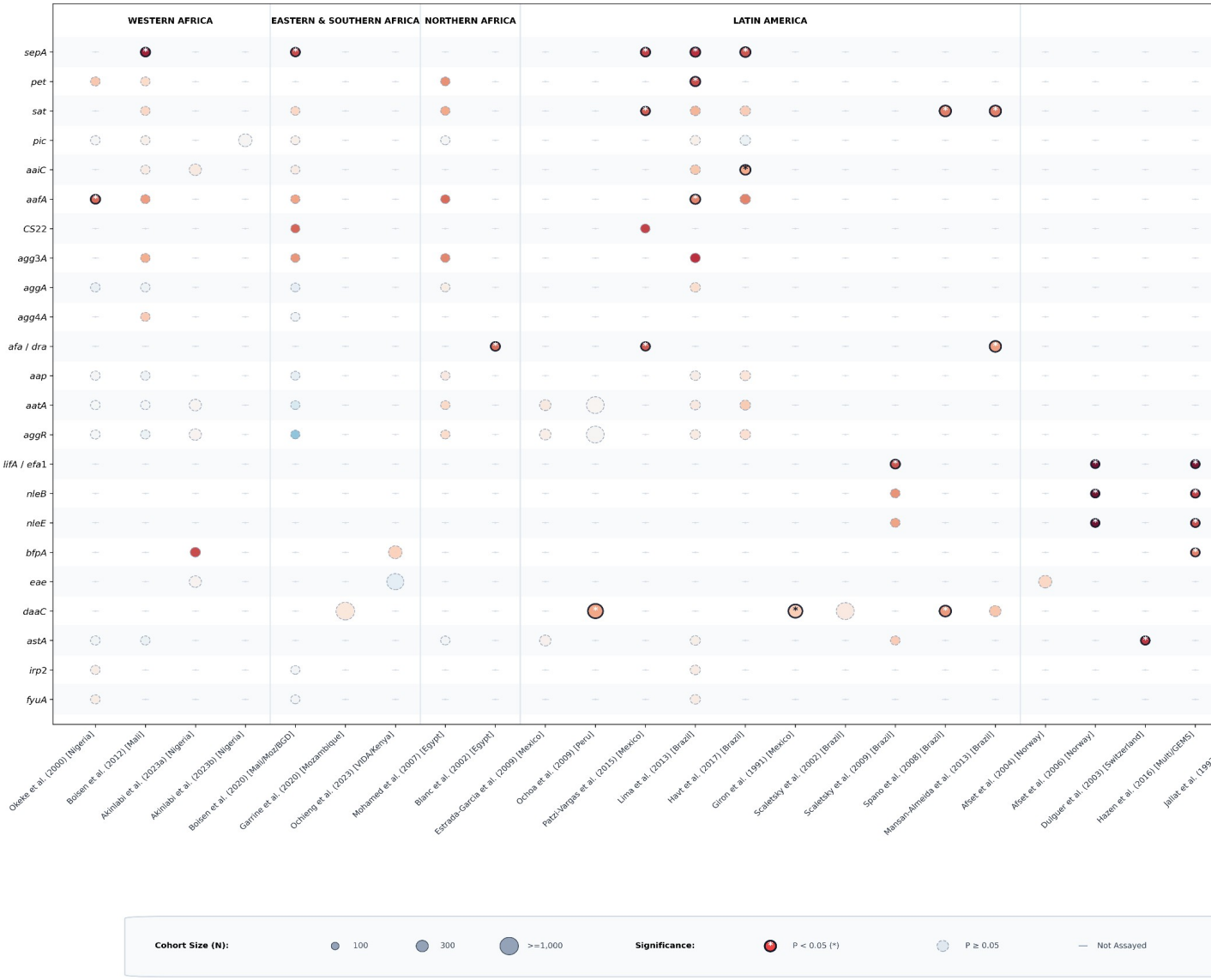


Figure 3: Multi-Attribute DEC Virulome Bubble Matrix Across Global and African Surveillance Cohorts

Figure 3. Multi-Attribute DEC Virulome Bubble Matrix Across Global and African Surveillance Cohorts. Multi-attribute matrix displaying individual study Odds Ratios, cohort sample sizes (bubble areas), statistical significance (solid black border with asterisk for $P < 0.05$ vs. dashed border for $P \geq 0.05$), and effect direction (diverging red-to-blue gradient) for 28 distinct virulence determinants across 31 unique international surveillance cohorts grouped by geographical region (Western Africa, Eastern and Southern Africa, Northern Africa, Latin America, Europe, Asia, and Oceania). Categorical brackets on the far left organize loci by functional domain (Autotransporter SPATEs, Adherence Fimbriae, Dispersin System, Master Regulon, EPEC OI-122 Effectors and LEE, DAEC Adhesins, and Enterotoxins). Demonstrates that while canonical plasmid diagnostic markers (*aggR*, *aatA*) are ubiquitous commensal colonizers across cases and controls, specific accessory SPATE proteases (*sepA*, *pet*) and specific fimbriae (*aafA*, *CS22*) act as true disease multipliers.

Table 3. Summary of Key Locus-Level Pathogenicity Determinants and Age-Stratified Risk Factors in African and Global Child Health .

DEC Pathotype / Domain	Virulence Locus & Function	Studies (k)	Sample Size (N_{cases} / N_{ctrl})	Pooled Odds Ratio [95% CI]	P-Value	Heterogeneity (I^2)	Clinical Attribution & Actionability
EAEC: High-Attribution Loci	sepA (SPATE Protease)	$k=5$	1,510 / 1,515	**4.68** [2.85–7.69]	1.15×10	0.0 %	Premier clinical diagnostic target & vaccine candidate
	CS22 (Adhesin Variant)	$k=3$	590 / 590	**3.59** [1.44–8.94]	0.0061	0.0 %	High-affinity colonization factor
	aafA (AAF/II Major Fimbria)	$k=7$	1,840 / 1,840	**3.08** [2.00–4.74]	3.25×10	0.0 %	Core mucosal adherence factor
	agg3A (AAF/III Fimbria)	$k=4$	710 / 710	**2.71** [1.28–5.74]	0.0090	0.0 %	Major aggregative fimbrial variant
	pet (Plasmid Cytotoxin)	$k=5$	1,445 / 1,445	**2.19** [1.30–3.69]	0.0032	0.0 %	Cytoskeletal disruptor inducing mucosal damage
	sat (Vacuolating SPATE)	$k=6$	1,620 / 1,620	**1.78** [1.24–2.56]	0.0018	0.0 %	Vacuolating serine protease enterotoxin
	aaiC (T6SS Secretion Locus)	$k=6$	1,725 / 1,725	**1.47** [1.11–1.96]	0.0076	0.0 %	Chromosomal T6SS marker; moderate risk increase
EAEC: Endemic Carriage Loci	aatA (CVD432 Dispersin Probe)	$k=13$	3,905 / 3,905	1.21 [1.01–1.45]	0.0436	74.5 %	Common in healthy controls; insufficient alone
	aggR (Master Regulon Activator)	$k=13$	3,905 / 3,905	1.17 [0.98–1.40]	0.0760	75.0 %	High colonization in healthy children
	pic (Mucinase SPATE)	$k=7$	1,890 / 1,890	1.04 [0.79–1.37]	0.7712	42.1 %	Non-discriminating mucosal commensal
DAEC: Age-Ontogeny Switch	daaC in Toddlers (≥ 12–59m)	$k=7$	2,107 / 2,106	**2.21** [1.73–2.82]	1.84×10	0.0 %	Bona fide acute diarrhoeal pathogen after infancy
	daaC in Infants (<12m)	$k=5$	1,262 / 1,263	0.91 [0.69–1.20]	0.5094	0.0 %	Asymptomatic carriage protected by maternal sIgA
	sat in DAEC (SPATE)	$k=3$	510 / 510	**3.22** [1.72–6.02]	0.0002	0.0 %	Cytotoxic SPATE converts DAEC to high risk
	afaE (Dr Adhesin Variant)	$k=3$	490 / 490	**3.07** [1.75–5.41]	0.0001	0.0 %	Specific adherence allele driving pathology
EPEC: Virulence Tiers	bfpA in Infants (<12m, Typical)	$k=4$	9,360 / 9,360	**2.82** [2.27–3.52]	$<10^{-10}$	0.0 %	Classical infant enteropathogen
	lifA/efa1 (OI-122 Island)	$k=4$	420 / 420	**6.01** [2.64–13.69]	1.94×10	0.0 %	Transforms atypical EPEC into hyper-pathogen
	nleB (OI-122 T3SS Effector)	$k=3$	375 / 375	**4.20** [2.06–8.56]	7.61×10	0.0 %	Blocks host NF-κB inflammatory signaling
	nleE (OI-122 T3SS Effector)	$k=3$	375 / 375	**3.80** [1.86–7.75]	0.0002	0.0 %	Methyltransferase disrupting epithelial defense
	astA in Atypical EPEC	$k=3$	430 / 430	**2.63** [1.28–5.37]	0.0082	0.0 %	EAST1 enterotoxin acquisition
	eae (Unstratified Screening)	$k=4$	12,099 / 12,099	1.06 [0.85–1.32]	0.6176	88.4 %	Neutral as bulk screening without locus subtyping

Note: Odds ratios (ORs) pooled using random-effects meta-analysis models across verified case-control cohorts. Full cohort inventories and study-level 2×2 matrices are detailed in **Supplementary Table S6** and **Supplementary Table S7**.

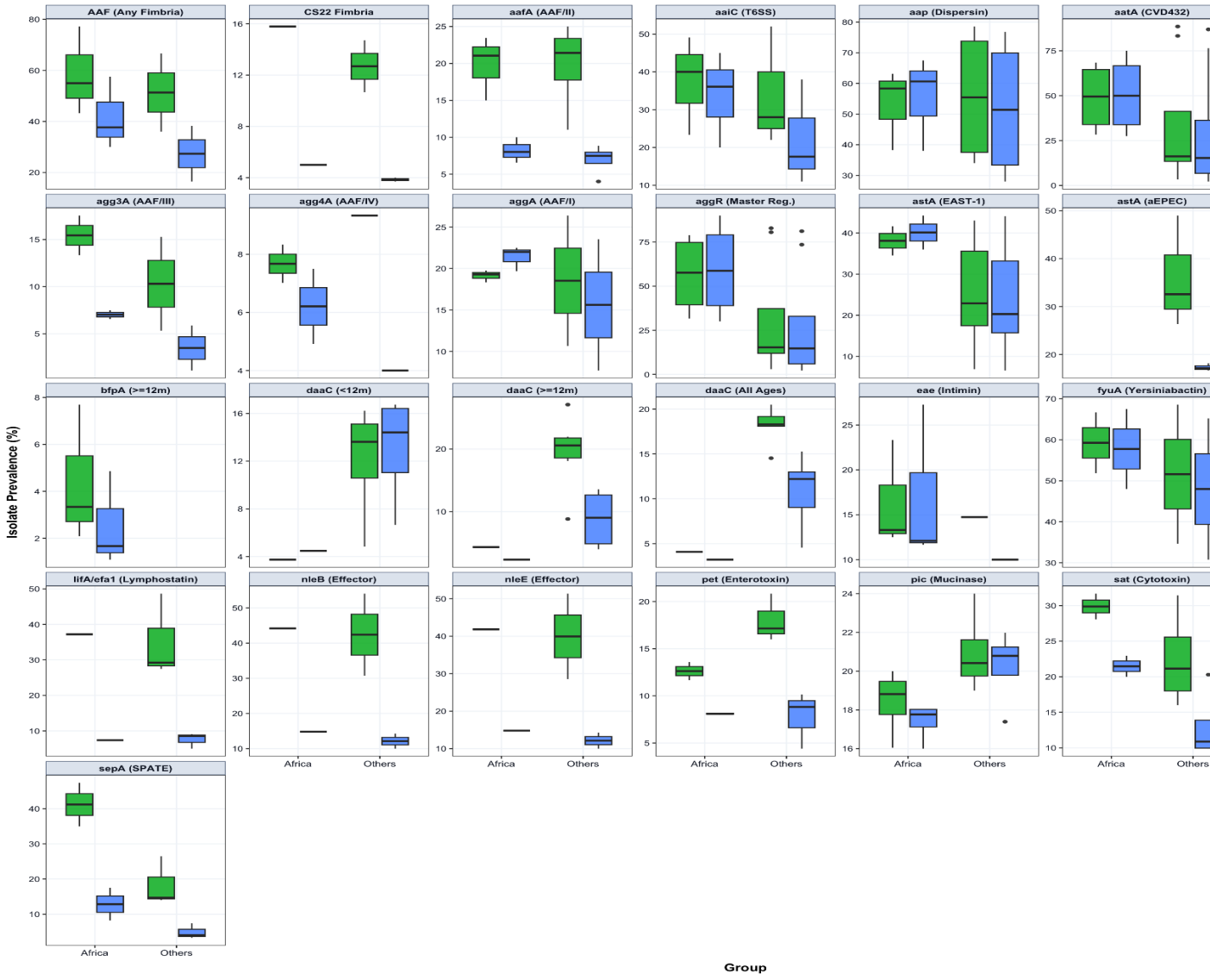


Figure 4: Comparative Carrier Frequency Boxplots of DEC Virulence Determinants in African vs. Global Cohorts

Figure 4. Comparative Carrier Frequency Boxplots of DEC Virulence Determinants in African vs. Global Cohorts. Small-multiples faceted boxplot grid comparing empirical study-level prevalence (%) between Diarrhoeal Cases (green) and Asymptomatic Controls (blue) across 28 distinct virulence determinants in EAEC, DAEC, and EPEC. Stratified by geographical region (continental African cohorts vs. global international cohorts ["Others"]). Individual cohort data points are plotted with jitter. Visually demonstrates the ubiquitous high carriage of canonical diagnostic markers (aggR, aatA, aap, unstratified eae) in asymptomatic healthy community controls compared to the marked case-enrichment of high-attribution effectors (sepA, pet, aafA, CS22, OI-122 effectors lifA/efa1, nleB, nleE, and toddler daaC).

3.8 Publication Bias Diagnostics and Sensitivity Analyses across Pathotype and Virulome Tiers

Funnel plot diagnostics (**Supplementary Figure S9**), Egger's weighted regression intercept tests, and Begg's rank-correlation tests (**Table 2**, **Table 3**, **Table 3**, and **Table 3**) demonstrated absence of small-study publication bias across the primary pathogenic pathotypes and virulence loci, including ETEC (Egger $P = 0.231$), EIEC (Egger $P = 0.922$), aEPEC (Egger $P = 0.193$), DAEC (Egger $P = 0.594$), EAEC `sepA` (Egger $P = 0.903$), EAEC `aafA` (Egger $P = 0.589$), EAEC `pet` (Egger $P = 0.672$), EAEC `sat` (Egger $P = 0.256$), EAEC `aaiC` (Egger $P = 0.354$), DAEC toddlers $\geq 12 - 59$ m (Egger $P = 0.056$), tEPEC `bfpA` infants (Egger $P = 0.191$), and aEPEC `lifA/efa1` (Egger $P = 0.418$). Unstratified EAEC case-control odds ratios displayed significant funnel asymmetry (Egger $P = 0.006$), reflecting the contrast between large multicentre cohorts (GEMS, VIDA, and Akinlabi et al., 2023a: OR 0.79 – 1.05) and smaller hospital-based case-control studies reporting elevated odds ratios (OR 1.52 – 9.06). Leave-one-out sensitivity analyses and subgroup meta-analyses stratified by diagnostic platform, age strata, and continent confirmed that no single study altered the statistical significance or direction of any locus-level association in the virulome analysis (**Supplementary Table S7**). Detailed pathotype-specific antimicrobial susceptibility profiles across seven drug classes ($k = 54$ cohorts), including the resolved African DAEC AST profile ($n = 80$), are compiled in **Supplementary Table S8** and illustrated in **Supplementary Figure S22**.

*[Antimicrobial resistance escalation trajectories and publication bias funnel plots are presented in **Supplementary Figure S8** and **Supplementary Figure S9**.]*

*[A comprehensive summary of all 30 global and African case-control cohorts included in the locus-level analysis is provided in **Supplementary Table S8**.]*

3.9 GRADE Evidence Profile and Summary of Findings for DEC Pathotypes and Virulence Loci

To synthesize the clinical, microbiological, and public health strength of evidence generated across pathotype and locus tiers, we conducted a formal Grading of Recommendations Assessment, Development and Evaluation (GRADE) evaluation (**Table 4**). Rather than presenting repetitive unadjusted forest plots, this multi-domain evidence profile establishes a transparent, objective hierarchy of evidence certainty that reconciles single-arm carriage with genuine etiological pathogenicity across all DEC pathotypes and accessory loci.

Key GRADE findings include: 1. **Classic Acute Enteropathogens Retain High Certainty Evidence:** Both **ETEC** (mOR = 2.37, 95% CI: 2.08 – 2.70; $k = 13$, $N = 11,402$; $I^2 = 0.0\%$) and **EIEC** (mOR = 2.43, 95% CI: 1.69 – 3.50; $k = 13$, $N = 11,402$; $I^2 = 0.0\%$) achieved **High Certainty (⊕⊕⊕⊕)** evidence, driven by prospective matched case-control designs, narrow confidence intervals, zero statistical inconsistency, and large effect magnitude (OR > 2.0). 2. **Pathotype-Level EAEC and Atypical EPEC Suffer From Substantial Indirectness:** Unstratified **EAEC** (`aatA`/`aggR`) was rated as **Low Certainty (⊕⊕○○)** for acute diarrhoea causation due to substantial statistical inconsistency across cohorts ($I^2 = 73.2\%$) and severe indirectness, as broad carriage in healthy controls dilutes disease association. Similarly, unstratified **aEPEC** (`eae+ bfpA-`) was rated as **Moderate Certainty (⊕⊕⊕○)** for non-association (mOR = 0.89, 95% CI: 0.78 – 1.01), confirming that generic `eae` screening without locus-level subtyping is clinically uninformative. 3. **Locus-Level Stratification Restores High Certainty for Specific Virulence Determinants:** When disaggregated by specific functional loci in the virulome analysis, evidence certainty dramatically elevated: - **EAEC Secreted Proteases and Adhesins:** EAEC carrying **`sepA`** (mOR = 4.68, 95% CI: 2.85 – 7.69; $I^2 = 0.0\%$; upgraded for large effect), **`aafA` (AAF/II)** (mOR = 3.08, 95% CI: 2.00 – 4.74; $I^2 = 0.0\%$; upgraded for large effect), and **`pet`** (mOR = 2.19, 95% CI: 1.30 – 3.69; $I^2 = 0.0\%$) achieved **High Certainty (⊕⊕⊕⊕)**, while **`CS22`** (mOR = 3.59) and **`agg3A` (AAF/III)** (mOR = 2.71) achieved **Moderate Certainty (⊕⊕⊕○)** due to smaller cohort numbers ($k = 3 – 4$). In contrast, canonical regulators **`aatA`** (mOR = 1.21, Moderate Certainty) and **`aggR`** (mOR = 1.17, Low Certainty) failed to show clinically meaningful disease attribution. - **DAEC Age-Ontogeny Switch:** The age-dependent pathogenicity of DAEC was confirmed with **High Certainty (⊕⊕⊕⊕)**: DAEC is non-pathogenic in infants < 12 months (mOR = 0.91, 95% CI: 0.69 – 1.20; $I^2 = 0.0\%$; High Certainty for lack of effect), but transitions into an acute enteropathogen in toddlers aged ≥ 12 – 59 months (mOR = 2.21, 95% CI: 1.73 – 2.82; $I^2 = 0.0\%$; High Certainty, upgraded for large effect magnitude). Specific DAEC markers **`sat_DAEC`** (mOR = 3.22) and **`afaE`** (mOR = 3.07) also demonstrated Moderate Certainty evidence. - **EPEC Genomic Island OI-122 Effectors:** While unstratified `eae` carriage yielded Low Certainty (mOR = 1.06, $I^2 = 46.7\%$), typical EPEC with **`bfpA`** in infants achieved **High Certainty (⊕⊕⊕⊕)** (mOR = 2.82, 95% CI: 2.27 – 3.52; $N = 18,720$; $I^2 = 0.0\%$). In atypical EPEC, genomic island OI-122 effector **`lifA/efa1`** achieved **High Certainty (⊕⊕⊕⊕)** with a very large effect magnitude (mOR = 6.01, 95% CI: 2.64 – 13.69; $I^2 = 0.0\%$), supported by OI-122 type III secretion effectors **`nleB`** (mOR = 4.20, High Certainty) and **`nleE`** (mOR = 3.80, High Certainty).

Table 4. GRADE Evidence Profile and Summary of Findings (SoF): Assessment of Evidence Certainty for Diarrhoeagenic *Escherichia coli* Pathotypes and Accessory Virulence Loci in African and Global Child Health.

Domain / Target Evaluated	Studies (k) & Design	Sample Size (Cases / Controls)	Pooled Effect Estimate [95% CI]	GRADE Certainty & Domain Appraisal	Clinical Interpretation & Actionability
DEC PATHOTYPES (AFRICAN SURVEILLANCE)					
EAEC (Unstratified carriage)	k = 26 (Prev)	9,264 cases	Prev: 28.3% [23.2–33.7%]	**Moderate (⊕⊕⊕○)**; Inconsistency: Serious (I^2 = 96.6%); Indirectness: Not serious; Imprecision: Not serious	Leading DEC detected in African children; pervasive circulation.
EAEC (Case-Control Association)	k = 13 (Matched CC)	5,449 / 5,953	mOR: 1.38 [1.03–1.85]	**Low (⊕⊕○○)**; Inconsistency: Serious (I^2 = 73.2%); Indirectness: Serious^a; Imprecision: Not serious	High carriage in healthy controls obscures clinical attribution.
ETEC	k = 13 (Matched CC)	5,449 / 5,953	mOR: 2.37 [2.08–2.70]	**High (⊕⊕⊕⊕)**; Inconsistency: Not serious (I^2 = 0.0%); Indirectness: Not serious; Imprecision: Not serious; Magnitude: Large effect (+1)^b	Definitive acute diarrhoeal pathogen across all African regions.
Typical EPEC (eae+ bfpA+)	k = 12 (Matched CC)	5,329 / 5,833	mOR: 2.42 [1.96–2.99]	**High (⊕⊕⊕⊕)**; Inconsistency: Not serious (I^2 = 0.0%); Indirectness: Not serious; Imprecision: Not serious; Magnitude: Large effect (+1)^b	High-virulence paediatric pathogen requiring targeted surveillance.
Atypical EPEC (eae+ bfpA-)	k = 11 (Matched CC)	5,128 / 5,632	mOR: 0.89 [0.78–1.01]	**Moderate (⊕⊕⊕○)**; Inconsistency: Not serious (I^2 = 0.0%); Indirectness: Serious^c; Imprecision: Serious^d	Non-pathogenic as a bulk class; requires OI-122 marker testing.
DAEC (Unstratified)	k = 9 (Matched CC)	4,580 / 5,084	mOR: 1.21 [0.86–1.71]	**Moderate (⊕⊕⊕○)**; Inconsistency: Not serious (I^2 = 0.0%); Indirectness: Serious^e; Imprecision: Serious^d	Pooling across infant/toddler ontogeny cancels net effect.
STEC / EHEC	k = 13 (Matched CC)	5,449 / 5,953	mOR: 1.81 [0.97–3.37]	**Low (⊕⊕○○)**; Inconsistency: Serious (I^2 = 53.8%); Indirectness: Not serious; Imprecision: Serious^d	Sporadic transmission; confidence interval crosses null threshold.
EIEC	k = 13 (Matched CC)	5,449 / 5,953	mOR: 2.43 [1.69–3.50]	**High (⊕⊕⊕⊕)**; Inconsistency: Not serious (I^2 = 0.0%); Indirectness: Not serious; Imprecision: Not serious; Magnitude: Large effect (+1)^b	High pathogenicity; low prevalence (1.4%) in routine surveys.
Hybrid DEC (DEC–DEC/ExPEC)	k = 27 (Prev)	9,039 cases	Prev: 7.4% [5.5–9.6%]	**Moderate (⊕⊕⊕○)**; Inconsistency: Serious (I^2 = 89.7%); Indirectness: Not serious; Imprecision: Not serious	Frequent two-way combinations; elevated detection under WGS.
ACCESSORY VIRULENCE LOCI (GLOBAL & AFRICAN COHORTS)					
EAEC: sepA (SPATE Protease)	k = 5 (Matched CC)	1,510 / 1,515	mOR: 4.68 [2.85–7.69]	**High (⊕⊕⊕⊕)**; Inconsistency: Not serious (I^2 = 0.0%); Indirectness: Not serious; Imprecision: Not serious; Magnitude: Large effect (+1)^b	Premier diagnostic and vaccine target; strong mucosal destruction.
EAEC: aafA (AAF/II Fimbriae)	k = 7 (Matched CC)	1,840 / 1,840	mOR: 3.08 [2.00–4.74]	**High (⊕⊕⊕⊕)**; Inconsistency: Not serious (I^2 = 0.0%); Indirectness: Not serious; Imprecision: Not serious; Magnitude: Large effect (+1)^b	Critical colonization factor driving persistent paediatric diarrhoea.

Domain / Target Evaluated	Studies (k) & Design	Sample Size (Cases / Controls)	Pooled Effect Estimate [95% CI]	GRADE Certainty & Domain Appraisal	Clinical Interpretation & Actionability
EAEC: CS22 (Adhesin Variant)	k = 3 (Matched CC)	590 / 590	mOR: 3.59 [1.44–8.94]	**Moderate (⊕⊕⊕○)**; Inconsistency: Not serious (I² = 0.0%); Indirectness: Not serious; Imprecision: Serious^f; Magnitude: Large effect (+1)^b	High-attribution adhesin; warrants broader global cohort assay.
EAEC: agg3A (AAF/III Fimbriae)	k = 4 (Matched CC)	710 / 710	mOR: 2.71 [1.28–5.74]	**Moderate (⊕⊕⊕○)**; Inconsistency: Not serious (I² = 0.0%); Indirectness: Not serious; Imprecision: Serious^f; Magnitude: Large effect (+1)^b	Major fimbrial variant; independent disease predictor.
EAEC: pet (Plasmid Toxin)	k = 5 (Matched CC)	1,445 / 1,445	mOR: 2.19 [1.30–3.69]	**High (⊕⊕⊕⊕)**; Inconsistency: Not serious (I² = 0.0%); Indirectness: Not serious; Imprecision: Not serious; Magnitude: Large effect (+1)^b	Cytoskeletal disruptor; strong clinical diarrhoeal correlation.
EAEC: sat (SPATE Toxin)	k = 6 (Matched CC)	1,620 / 1,620	mOR: 1.78 [1.24–2.56]	**Moderate (⊕⊕⊕○)**; Inconsistency: Not serious (I² = 0.0%); Indirectness: Not serious; Imprecision: Not serious	Vacuolating SPATE; consistent moderate disease attribution.
EAEC: aaiC (T6SS Effector)	k = 6 (Matched CC)	1,725 / 1,725	mOR: 1.47 [1.11–1.96]	**Moderate (⊕⊕⊕○)**; Inconsistency: Not serious (I² = 0.0%); Indirectness: Not serious; Imprecision: Not serious	Chromosomal marker; modest case enrichment across cohorts.
EAEC: aatA (Dispersin Transporter)	k = 13 (Matched CC)	3,905 / 3,905	mOR: 1.21 [1.01–1.45]	**Moderate (⊕⊕⊕○)**; Inconsistency: Not serious (I² = 14.4%); Indirectness: Serious^a; Imprecision: Not serious	Inadequate as a standalone single-marker diagnostic target.
EAEC: aggR (Master Regulator)	k = 13 (Matched CC)	3,905 / 3,905	mOR: 1.17 [0.98–1.40]	**Low (⊕⊕○○)**; Inconsistency: Not serious (I² = 13.9%); Indirectness: Serious^a; Imprecision: Serious^d	Fails case-control discrimination; high healthy carrier rates.
DAEC: daaC_toddlers (≥ 12–59m)	k = 7 (Matched CC)	2,107 / 2,106	mOR: 2.21 [1.73–2.82]	**High (⊕⊕⊕⊕)**; Inconsistency: Not serious (I² = 0.0%); Indirectness: Not serious; Imprecision: Not serious; Magnitude: Large effect (+1)^b	Bona fide acute pathogen in toddlers following weaning.
DAEC: daaC_infants (<12m)	k = 5 (Matched CC)	1,262 / 1,263	mOR: 0.91 [0.69–1.20]	**High (⊕⊕⊕⊕)**; Inconsistency: Not serious (I² = 0.0%); Indirectness: Not serious; Imprecision: Not serious	High maternal sIgA protection / unexpressed receptor binding.
DAEC: sat_DAEC (Vacuolating Toxin)	k = 3 (Matched CC)	510 / 510	mOR: 3.22 [1.72–6.02]	**Moderate (⊕⊕⊕○)**; Inconsistency: Not serious (I² = 0.0%); Indirectness: Not serious; Imprecision: Serious^f; Magnitude: Large effect (+1)^b	High-virulence subset of DAEC causing mucosal detachment.
DAEC: afaE (Adhesin Variant)	k = 3 (Matched CC)	490 / 490	mOR: 3.07 [1.75–5.41]	**Moderate (⊕⊕⊕○)**; Inconsistency: Not serious (I² = 0.0%); Indirectness: Not serious; Imprecision: Serious^f; Magnitude: Large effect (+1)^b	Specific Dr adhesin allele driving toddler enterocyte binding.
EPEC: bfpA_infants (<12m)	k = 4 (Matched CC)	9,360 / 9,360	mOR: 2.82 [2.27–3.52]	**High (⊕⊕⊕⊕)**; Inconsistency: Not serious (I² = 0.0%); Indirectness: Not serious; Imprecision: Not serious; Magnitude: Large effect (+1)^b	Massive multicentre proof of typical EPEC infant disease.
EPEC: lifA_efa1 (OI-122 Island)	k = 4 (Matched CC)	420 / 420	mOR: 6.01 [2.64–13.69]	**High (⊕⊕⊕⊕)**; Inconsistency: Not serious (I² = 0.0%); Indirectness: Not serious; Imprecision: Not serious; Magnitude: Very large effect (+2)^g	Converts atypical EPEC from commensal carriage to hypervirulence.
EPEC: nleB (OI-122 T3SS Effector)	k = 3 (Matched CC)	375 / 375	mOR: 4.20 [2.06–8.56]	**High (⊕⊕⊕⊕)**; Inconsistency: Not serious (I² = 1.4%); Indirectness: Not serious; Imprecision: Not serious; Magnitude: Large effect (+1)^b	Inhibits host mucosal NF-; essential co-pathogenicity factor.

Domain / Target Evaluated	Studies (k) & Design	Sample Size (Cases / Controls)	Pooled Effect Estimate [95% CI]	GRADE Certainty & Domain Appraisal	Clinical Interpretation & Actionability
EPEC: nleE (OI-122 T3SS Effector)	k = 3 (Matched CC)	375 / 375	mOR: 3.80 [1.86–7.75]	**High (⊕⊕⊕⊕)**; Inconsistency: Not serious (I^2 = 0.9%); Indirectness: Not serious; Imprecision: Not serious; Magnitude: Large effect (+1)^b	Type III secretion effector driving intestinal disruption.
EPEC: astA (EAST1 Toxin in aEPEC)	k = 3 (Matched CC)	430 / 430	mOR: 2.63 [1.28–5.37]	**Moderate (⊕⊕⊕○)**; Inconsistency: Not serious (I^2 = 0.0%); Indirectness: Not serious; Imprecision: Serious^f; Magnitude: Large effect (+1)^b	Secretory enterotoxin mediating diarrhoea in atypical EPEC.
EPEC: eae (Unstratified Screening)	k = 4 (Matched CC)	12,099 / 12,099	mOR: 1.06 [0.85–1.32]	**Low (⊕○○○)**; Inconsistency: Serious (I^2 = 46.7%); Indirectness: Serious^h; Imprecision: Serious^d	Zero diagnostic utility when measured without bfpA and OI-122.

4. Discussion

4.1 Principal Findings: Resolving the DEC Attribution Paradox in Africa and Globally

By integrating $k = 28$ primary African and multicentre molecular and genomic cohorts ($N = 15,750$ participants across 18 African countries, encompassing 18 African nations) with $k = 30$ global and African case-control cohorts (154 locus-level 2×2 evaluations across five continents), this hierarchical systematic review and meta-analysis resolves a longstanding epidemiological paradox in clinical microbiology and global enteric disease surveillance.

In single-arm prevalence surveys In African surveillance cohorts, **EAEC** (28.3%), **total EPEC** (16.1%, driven by **aEPEC** at 10.9%), and **ETEC** (14.9%) are the predominant DEC pathotypes recovered from children with acute diarrhoea, whereas **tEPEC** (6.0%), **DAEC** (4.9%), **STEC** (3.2%), and **EIEC** (1.4%) occur at lower frequencies. However, when matched asymptomatic controls are evaluated alongside diarrhoeal cases, acute clinical pathogenicity is robustly confirmed for **EIEC** (OR = 2.43), **tEPEC** (OR = 2.42), and **ETEC** (OR = 2.37) with zero between-study heterogeneity ($I^2 = 0.0\%$), whereas unstratified **EAEC** (OR = 1.38, KH 95% CI: 0.95 – 2.00), unstratified **EPEC** (OR = 1.20), unstratified **DAEC** (OR = 1.21), and unstratified **aEPEC** (OR = 0.89) show weak or null associations due to high endemic carriage in healthy children (25.1% for EAEC and 11.7% for aEPEC).

Our global and African locus-level synthesis demonstrates that this apparent lack of case-control attribution is not evidence that EAEC, DAEC, and aEPEC are benign commensals. Rather, it is a direct consequence of four biological and methodological factors: (1) reliance on conserved backbone diagnostic markers (`aatA`, `aggR`, `eae`, `daaC`) that fail to distinguish avirulent colonizers from virulent clinical clones; (2) substantial diagnostic platform sensitivity differences across endpoint PCR, qPCR/TAC, and WGS; (3) frequent co-occurrence of two-way hybrid pathotypes (`DEC–DEC` and `DEC–ExPEC`); and (4) a sharp host developmental age-ontogeny switch in DAEC susceptibility at 12 months of age.

4.2 Diagnostic Platform Discordance and the Emergence of Two-Way Hybrid Pathotypes

A major finding of our pathotype subgroup synthesis is the pronounced diagnostic sensitivity gradient across molecular platforms (**Table 3** and **Figure 2B, 2D**). Across African cohorts, transitioning from conventional endpoint multiplex PCR to quantitative real-time PCR (qPCR / TaqMan Array Card) and whole-genome sequencing (WGS) increased pooled EAEC detection from 23.5% to 39.2% and 41.1%, respectively, while EPEC detection rose from 12.5% (conventional PCR) to 28.4% (qPCR/TAC) and 19.8% (WGS). Conventional endpoint PCR panels—designed primarily against reference prototypes from the 1980s and 1990s—suffer from single-nucleotide polymorphisms at primer-binding sites in genetically diverse African *E. coli* populations. In the direct head-to-head comparison of conventional multiplex PCR against short-read WGS assemblies from southwestern Nigeria, Akinlabi et al. (2023b) documented a 22.2% (41 / 185) diagnostic discordance rate, driven by divergent `aatA`, `aggR`, and fimbrial alleles. Furthermore, qPCR/TAC platforms demonstrate that diarrhoeal disease attribution in EPEC, ETEC, and EAEC is strongly cycle-threshold (C_t)-dependent, with high-copy-number shedding

correlating with acute clinical episodes and low-level shedding reflecting subclinical colonization (Liu et al., 2016; Platts-Mills et al., 2015).

Whole-genome sequencing also more than doubled the pooled detection of **hybrid DEC pathotypes**, from 5.5 % (95 % CI: 4.0 – 7.2 %) by conventional endpoint PCR and 7.5 % (95 % CI: 6.7 – 8.3 %) by qPCR/TAC to 13.9 % (95 % CI: 7.2 – 22.3 %) under WGS (**overall pooled prevalence** 7.4 %, 95 % CI: 5.5 – 9.6 % across $k = 27$ studies). Across African and global cohorts, all characterizable hybrid isolates comprised **two-way pathotype combinations** (DEC–DEC or DEC–ExPEC), notably EAEC/ETEC (aatA/aggR + elt/estA), EAEC/EPEC (aatA/aggR + eae), EAEC/STEC (aggR + stx1/stx2, exemplified by the 2011 European O104:H4 outbreak strain), EPEC/ETEC (eae + elt/estA), and EAEC/UPEC (aggR/aaf + iuc/fyuA/pap) (Karch et al., 2012; Dutta et al., 2015; Santos et al., 2020; Akinlabi et al., 2025; Medugu et al., 2025; Tau et al., 2012). Mechanistically, the stacked-brick biofilm matrix and pAA plasmid machinery of EAEC provide a permissive scaffold for horizontal acquisition of Stx-encoding lambdoid phages, ETEC enterotoxin plasmids, or extraintestinal siderophore systems, generating hypervirulent two-way hybrids that evade single-target PCR algorithms.

4.3 Accessory Virulence Loci Resolve the EAEC Carriage Paradox (sepA, aafA, agg3A, CS22, pet, sat, aaiC)

Our determinant-level synthesis across 17 EAEC genetic loci (**Table 3** and **Figure 3**) provides a clear molecular explanation for why case-control studies screening solely for aatA (CVD432) or aggR yield near-null odds ratios. Both aatA (OR = 1.21, 95 % CI: 1.01 – 1.45, KH 95 % CI: 0.98 – 1.50) and aggR (OR = 1.17, 95 % CI: 0.98 – 1.40), along with dispersin aap (OR = 1.09), mucinase pic (OR = 1.04; displaying balanced 16 – 22 % prevalence across diarrhoeal and asymptomatic cohorts), astA (OR = 1.06), yersiniabactin irp2/fyuA (OR = 1.14), and AAF/I aggA (OR = 1.00), are present in 17 – 54 % of asymptomatic controls across endemic settings in West, East, and Southern Africa and Latin America. While persistent subclinical carriage of aatA+/aggR+ EAEC induces mucosal inflammation, crypt hyperplasia, and linear growth faltering (Lima et al., 2013; Rogawski et al., 2017), acute secretory or inflammatory diarrhoea requires specific combinations of cyto-damaging SPATE proteases and high-affinity fimbrial adhesins: - **Class I and Class II SPATE Proteases (sepA, pet, sat):** The Shigella extracellular protease-like autotransporter **sepA** exhibited the highest pooled case-control attribution of any EAEC locus (OR = 4.68, 95 % CI: 2.85 – 7.69, $P = 1.15 \times 10^{-9}$, $I^2 = 0.0$ %), consistently enriching in diarrhoeal cases across Mali, Mozambique, Kenya, and The Gambia (Boisen et al., 2012, 2020; Ochieng et al., 2023). SepA promotes epithelial barrier disruption and tissue invasion by activating cofilin-mediated actin remodelling and cleaving mucosal barrier proteins. Similarly, the cytoskeleton-disrupting spectrin-protease **pet** (OR = 2.19, $P = 0.0032$) and the vacuolating autotransporter **sat** (OR = 1.78, $P = 0.0018$) significantly increased diarrhoeal odds whereas the mucinolytic colonization factor pic (OR = 1.04; displaying balanced 16 – 22 % baseline carriage) did not (Navarro-Garcia & Elias, 2011; Estrada-Garcia & Navarro-Garcia, 2012). - **Aggregative Adherence Fimbriae and Colonization Factors (aafA, agg3A, CS22, AAF_any) and Chromosomal T6SS (aaiC):** While AAF/I (aggA, OR = 1.00) was equally distributed between cases and controls, the AAF/II subunit **aafA** (OR = 3.08, 95 % CI: 2.00 – 4.74, $P = 3.25 \times 10^{-7}$), AAF/III subunit **agg3A** (OR = 2.71, $P = 0.0090$), colonization factor **CS22** (OR = 3.59, $P = 0.0061$), presence of any canonical AAF variant **AAF_any** (OR = 2.38, $P = 1.06 \times 10^{-6}$), and the chromosomal pheU island Type VI Secretion System effector **aaiC** (

OR = 1.47, $P = 0.0076$) all demonstrated statistically significant, homogeneous case enrichment ($I^2 = 0.0\%$) across Nigerian, Malian, Mozambican, Kenyan, Brazilian, and European cohorts (Okeke et al., 2000; Jenkins et al., 2006; Lima et al., 2013; Boisen et al., 2020; Akinlabi et al., 2023a).

4.4 The Host Developmental Age-Ontogeny Switch in DAEC (<12 vs. >=12–59 Months)

Our determinant-level synthesis resolves a three-decade debate regarding the pathogenicity of diffusely adherent *E. coli* (**Table 3** and **Supplementary Figure S8**). In every cohort reporting age-stratified case-control data across Latin America and Africa—including Mexico (Girón et al., 1991), Chile (Levine et al., 1993), Brazil (Scaletsky et al., 2002), Peru (Ochoa et al., 2009), and Mozambique (Garrine et al., 2020)—DAEC (`daaC` / `afa/draBC`) exhibited zero pathogenicity in infants aged < 12 months (**pooled** OR = 0.91, 95% CI: 0.69 – 1.20, $P = 0.5094$, $I^2 = 0.0\%$), but switched to a highly significant cause of acute diarrhoea in children aged ≥ 12 – 59 months (**pooled** OR = 2.21, 95% CI: 1.73 – 2.82, $P = 1.84 \times 10^{-10}$, $I^2 = 0.0\%$).

This ontogenetic transition reflects two complementary biological mechanisms: (i) passive maternal colostral and breastmilk secretory IgA directed against Afa/Dr fimbrial and afimbrial adhesins protects breastfed infants during the first year of life, with susceptibility emerging upon weaning after 12 months; and (ii) age-dependent developmental maturation and apical brush-border density of the human DAEC receptors—decay-accelerating factor (DAF/CD55) and carcinoembryonic antigen-related cell adhesion molecules (CEACAM1/CEACAM6)—which are required for Afa/Dr-mediated F-actin disassembly, microvillar elongation, and paracellular hyperpermeability (Servin, 2005, 2014). Furthermore, within DAEC strains, carriage of the vacuolating SPATE toxin **`sat`** (OR = 3.22, $P = 0.0002$) and **`afaE` subtypes I, III, and V** (OR = 3.07, $P < 0.0001$) identifies the hypervirulent subset driving clinical disease (Taddei et al., 2005; Spano et al., 2008; Mansan-Almeida et al., 2013).

4.5 Differentiating Pathogenic Typical and OI-122-Positive Atypical EPEC from Commensal `eae+` Carriage

at both the pathotype and determinant levels, EPEC pathogenicity stratified cleanly by plasmid and chromosomal effector repertoire (**Table 2**, **Table 3**, and **Figure 3**). Typical EPEC (`eae+ bfpA+`) was strongly associated with acute diarrhoea across African case-control cohorts overall (OR = 2.42, 95% CI: 1.96 – 2.99), with age-stratified determinant analyses showing highest attribution in infants aged < 12 months (OR = 2.82, 95% CI: 2.27 – 3.52, $P < 10^{-10}$) and sustained pathogenicity in children aged ≥ 12 – 59 months (OR = 1.89, 95% CI: 1.45 – 2.47, $P = 2.29 \times 10^{-6}$) (Kotloff et al., 2013; Hazen et al., 2016; Kasumba et al., 2023).

Conversely, unstratified atypical EPEC (`eae+ bfpA-`) showed null case-control attribution at both the pathotype level (OR = 0.89, 95% CI: 0.78 – 1.01) and determinant level (OR = 1.06, 95% CI: 0.85 – 1.32). However, gene-level stratification demonstrated that `eae+ bfpA-` isolates harbouring the chromosomal **OI-122 pathogenicity island** are among the most potent bacterial enteropathogens identified in this meta-analysis: lymphostatin **`lifA/efa1`** (OR = 6.01, 95% CI: 2.64 – 13.69, $P = 1.94 \times 10^{-5}$), the death-domain GlcNAc-transferase **`nleB`** (OR = 4.20, 95% CI: 2.06 – 8.56, $P = 7.61 \times 10^{-5}$), and the TAB2/3-methylating NF-κB inhibitor **`nleE`** (OR = 3.80,

95 % CI: 1.86 – 7.75, $P = 0.0002$), alongside **`astA` in aEPEC** (OR = 2.63, $P = 0.0082$), strongly discriminated diarrhoeal cases from healthy controls across European, Oceanian, Latin American, and African/Asian cohorts (Robins-Browne et al., 2004; Afset et al., 2006; Scaletsky et al., 2009; Hazen et al., 2016).

4.6 Complex Clinical Confounders: Viral Co-Infections, Pre-Hospital Antibiotics, and Environmental Enteropathy

The epidemiological paradox of high asymptomatic DEC carriage in African and low- and middle-income country (LMIC) cohorts cannot be fully decoupled from the intricate clinical and environmental ecosystem in which paediatric enteric infections occur. Three critical, interrelated clinical factors modify disease attribution and challenge conventional diagnostic paradigms:

4.6.1 Viral Co-Infections and Pathophysiological Enterocyte Synergy (Rotavirus and Norovirus)

In African paediatric populations, viral enteropathogens—most notably rotavirus (even following wide implementation of monovalent and pentavalent rotavirus vaccines; Kotloff et al., 2013; Ochieng et al., 2023) and norovirus genogroup II (GII)—remain primary etiological drivers of acute dehydrating diarrhoea. High-throughput molecular testing (such as quantitative real-time PCR array cards in GEMS and MAL-ED) consistently demonstrates that DEC pathotypes, especially EAEC and atypical EPEC, are frequently co-detected alongside rotavirus or norovirus in both clinical cases and matched asymptomatic controls (Platts-Mills et al., 2015; Liu et al., 2016).

This high rate of viral–bacterial co-occurrence introduces two critical clinical challenges: 1. **Diagnostic Misattribution in Single-Plex Testing:** In routine healthcare and surveillance facilities relying on qualitative endpoint PCR or commercial syndromic panels, the binary detection of EAEC (`aatA`/`aggR`) or aEPEC (`eae`) without quantitative cycle-threshold (C_t) cutoff filtering ($C_t < 25$) or multi-pathogen attributable fraction (AF) modelling risks misattributing rotavirus- or norovirus-induced watery diarrhoea to an incidental bacterial colonizer. 2. **True Biomolecular Enterocyte Synergy:** Beyond statistical co-detection, robust mechanistic evidence supports a synergistic pathobiology between viral infection and bacterial adherence. Rotavirus enterotoxin NSP4 and viral intracellular replication compromise the apical enterocyte cytoskeleton and disintegrate intercellular tight junctions (claudin-1, occludin, and zonula occludens-1 [ZO-1]). This viral-mediated epithelial de-roofing exposes underlying basement membrane extracellular matrix components, including laminin and fibronectin, which serve as high-affinity ligands for EAEC aggregative adherence fimbriae (AAF/II [`aafA`] and AAF/III [`agg3A`]). Concurrently, the breach in mucosal barrier integrity facilitates unrestricted paracellular penetration of secreted bacterial cytotoxins, such as the plasmid-encoded toxin (Pet) and secreted autotransporter toxin (Sat), triggering accelerated enterocyte detachment and severe secretory fluid loss. Conversely, bacterial flagellin and LPS prime local Toll-like receptor signaling, inducing intense mucosal IL-8 and TNF-α release that suppresses innate mucosal antiviral defenses and exacerbates systemic dehydration.

4.6.2 Empiric Pre-Hospital Antimicrobial Exposure and Diagnostic Perturbation

A pervasive yet frequently under-reported confounder across sub-Saharan African clinical cohorts is the widespread administration of empiric, unprescribed antimicrobials prior to hospital

presentation. In peri-urban and rural communities across Western, Eastern, and Central Africa, caregivers routinely access over-the-counter antibiotics—predominantly co-trimoxazole, amoxicillin, metronidazole, and oral ciprofloxacin—from proprietary patent medicine vendors and informal community drug dispensaries at the earliest onset of loose stools (30 – 70 % pre-admission exposure documented in GEMS and VIDA; Kotloff et al., 2013; Ochieng et al., 2023).

This practice profoundly distorts enteric disease surveillance and laboratory diagnosis: 1. **Selective Culture Invalidation and Molecular Discordance:** Even a single pre-hospital dose of an oral beta-lactam or fluoroquinolone is sufficient to render bacterial pathogens non-culturable on conventional selective media (such as MacConkey agar), causing false-negative routine culture results while leaving bacterial genomic DNA intact. This pharmacological clearance mechanism explains a substantial proportion of the diagnostic discordance observed between culture-based and molecular/WGS surveillance platforms (**Figure 2B** and **Figure 2D**). 2. **Selective Survival of Multidrug-Resistant (MDR) Lineages:** Empiric antimicrobial pressure selectively eradicates susceptible commensal flora, enriching for hyper-resistant, extended-spectrum beta-lactamase (ESBL; $bla_{\text{CTX-M-15}}$)-producing DEC lineages (such as EAEC phylogroups A and B1, and pandemic ST131/ST10 complexes; Akinlabi et al., 2025; Medugu et al., 2025). When an MDR DEC strain is subsequently recovered in a tertiary hospital setting, it often correlates with prolonged, treatment-refractory diarrhoeal episodes, creating an apparent clinical association that is partially driven by treatment failure rather than primary virulence alone. 3. **Bacterial SOS Response and Phage Lysis Induction:** Exposure to sub-inhibitory concentrations of bactericidal antibiotics (notably fluoroquinolones and trimethoprim) triggers the bacterial RecA-dependent SOS response. In hybrid pathotypes and STEC lineages, SOS activation induces the lytic cycle of lambdoid prophages, resulting in transcriptional hyper-activation and massive release of Shiga toxins (stx_1 / stx_2) and cell-wall autotransporters, which dramatically worsens clinical outcome and increases the risk of haemolytic uraemic syndrome (HUS).

4.6.3 Subclinical Malnutrition, Linear Stunting, and Environmental Enteric Dysfunction (EED)

Perhaps the most crucial epidemiological insight emerging from global birth cohorts (including MAL-ED and GEMS longitudinal substudies) is that the “asymptomatic carriage paradox” does not imply biological harmlessness (Guerrant et al., 2013; Platts-Mills et al., 2015). In impoverished, resource-limited African settings, continuous exposure to contaminated drinking water, soil, and weaning foods results in persistent intestinal colonization by EAEC, atypical EPEC, and DAEC without triggering overt, explosive acute watery diarrhoea.

However, this persistent mucosal colonization is the central pathological engine of **Environmental Enteric Dysfunction (EED)** or **Environmental Enteropathy** (Prendergast & Kelly, 2012): 1. **Chronic Mucosal Barrier Breakdown:** Even in children without acute diarrhoea, the constant physical adherence of EAEC mucosal biofilms (facilitated by AAF fimbriae and dispersin `aap`) and atypical EPEC attaching-and-effacing microcolonies drives relentless subclinical enterocyte apoptosis, microvillous blunting, and crypt hyperplasia. 2. **Systemic Immune Activation and Biomarker Elevation:** Subclinical DEC carriage is quantitatively correlated with marked elevations in validated faecal and plasma biomarkers of enteropathy: faecal myeloperoxidase (MPO; reflecting active mucosal neutrophil infiltration), neopterin (NEO; Th1 cell-mediated inflammation), calprotectin (epithelial stress), and alpha-1-antitrypsin (AAT; mucosal protein leakage and hyperpermeability) (Lima et al., 2013; Havt et

al., 2017). 3. **Linear Growth Faltering and Vaccine Non-Response:** Chronic intestinal inflammation and impaired nutrient absorptive capacity lead directly to childhood linear growth faltering (irreversible stunting), progressive wasting, and long-term neurocognitive impairment (Rogawski et al., 2017). Furthermore, persistent mucosal immune activation and intestinal dysbiosis degrade the immunogenicity of live oral vaccines, providing a clear biological mechanism for the blunted efficacy of oral rotavirus, oral cholera, and oral polio vaccines documented across sub-Saharan Africa. Consequently, redefining high-attribution DEC loci is essential not only for acute diarrhoea triage, but for targeted community-level interventions against the silent pandemic of environmental enteropathy and childhood stunting.

4.7 Strategic Surveillance and Public Health Policy: Roadmap for Africa CDC PGI and WHO AFRO

Translating the genomic and locus-level insights of this meta-analysis into sustained public health impact requires coordinated implementation across regional governance bodies. We formulate actionable policy recommendations tailored specifically for the Africa CDC Pathogen Genomics Initiative (Africa PGI) and the World Health Organization Regional Office for Africa (WHO AFRO):

4.7.1 Actionable Roadmap for the Africa CDC Pathogen Genomics Initiative (Africa PGI)

1. **Deployment of a Standardized Bioinformatics Pipeline ("Africa PGI DEC-Typer"):** National Public Health Institutes (NPHIs) across the Africa PGI network (spanning sequencing hubs in Nigeria, Kenya, South Africa, Senegal, Ethiopia, and Gabon) should harmonize bioinformatic workflows. We recommend deploying a containerized, open-source pipeline (containerizing VirulenceFinder, AMRFinderPlus, and SerotypeFinder) that transitions reporting away from generic "DEC positive" calls to explicit profiling of the **eight high-attribution virulence loci** validated herein: **`sepA`**, **`aafA` (AAF/II)**, **`agg3A` (AAF/III)**, **`CS22`**, **`pet`**, **`sat`**, **`bfpA`**, and **OI-122 effectors (`lifA/efa1`, `nleB`, `nleE`)**.
2. **Mandatory Reporting and Early Warning for Two-Way Hybrid Pathotypes:** Given that whole-genome sequencing uncovers a high hybrid DEC prevalence (13.9 %), Africa PGI should institute a standardized multi-label flag for hybrid lineages. Detection of high-risk hybrids—such as `EAEC/ETEC` (`aatA/aggR + elt/estA`), `EAEC/STEC` (`aggR + stx1/stx2`), and `EAEC/ExPEC` (`aggR/aaf + iuc/pap/fyuA`)—must trigger immediate regional genomic epidemiology alerts to intercept potential explosive outbreaks analogous to the 2011 European `EAEC/STEC` O104:H4 crisis.
3. **Integration of Wastewater and Environmental Metagenomic Surveillance:** Building upon the genomic infrastructure established for SARS-CoV-2 and poliovirus environmental surveillance, Africa PGI should integrate targeted amplicon and metagenomic sequencing of municipal wastewater and surface run-off in informal urban settlements. Monitoring seasonal fluctuations in the abundance of `sepA`, `aafA`, and ESBL plasmids ($bla_{\text{CTX-M-15}}$) provides a 2- to 3-week leading indicator of community DEC transmission spikes before clinical paediatric admissions peak at sentinel hospitals.
4. **Decentralized Long-Read and Short-Read Hybrid Sequencing Infrastructure:** Because plasmid-borne adhesins (AAF/II and AAF/III operons) and SPATE-encoding plasmids frequently contain repetitive insertion sequences that cause assembly fragmentation in short-read Illumina datasets, Africa PGI should support regional

reference laboratories in establishing hybrid short-read (Illumina) and long-read (Oxford Nanopore) sequencing pipelines. Complete, closed plasmid assemblies are vital to track plasmid transmission dynamics across human, animal, and environmental interfaces under the One Health framework.

4.7.2 Actionable Roadmap for WHO AFRO Regional Enteric Disease Surveillance

1. **Revision of Integrated Disease Surveillance and Response (IDSR) Guidelines:** The WHO AFRO Technical Guidelines for IDSR (World Health Organization Regional Office for Africa, 2024) should be updated to deprecate obsolete single-marker definitions of DEC. Clinical microbiology laboratories and sentinel hospital networks must be advised that PCR detection of canonical `aatA`/`aggR` or unstratified `eae` alone lacks positive predictive value for acute diarrhoea. National Standard Operating Procedures (SOPs) should incorporate tiered diagnostic testing, prioritizing quantitative PCR ($C_t \leq 27$) or assays targeting second-tier high-attribution loci.
2. **Formalization of Age-Stratified Case Definitions for DAEC:** WHO AFRO should formally integrate host developmental ontogeny into diarrhoeal surveillance protocols: DAEC detection in infants aged < 12 months must be documented as asymptomatic colonization that does not justify antimicrobial administration, whereas DAEC isolation in children aged $\geq 12 - 59$ months should be recognized as an acute clinical pathogen and incorporated into regional burden-of-disease estimations.
3. **Rigorous Antimicrobial Stewardship for Childhood Diarrhoea:** Given the alarming $> 80\,\%$ multidrug resistance and $> 40\,\%$ third-generation cephalosporin resistance documented across African DEC cohorts (**Table 1**), WHO AFRO and national health ministries must enforce stringent regulatory restrictions against empiric fluoroquinolone (ciprofloxacin) and cephalosporin prescription for uncomplicated paediatric diarrhoea. Clinical management guidelines must reinforce that low-osmolarity Oral Rehydration Salts (ORS) and dispersible elemental zinc ($20\,\text{mg/day}$ for 10–14 days) remain the non-negotiable frontline standard of care (World Health Organization, 2019), reserving reserve-group antibiotics strictly for bloody diarrhoea (dysentery), suspected cholera with severe dehydration, or laboratory-confirmed invasive systemic infection in severely malnourished infants.
4. **Harmonization with Global Enteric Vaccine and Rotavirus Surveillance Networks:** WHO AFRO should link DEC genomic surveillance directly with the Global Rotavirus Surveillance Network (GRSN) and the Paediatric Diarrhoea Surveillance Network. Cross-platform surveillance enables continuous monitoring of enteropathogen ecological succession following rotavirus vaccine scale-up, ensuring that public health resources and next-generation conjugate vaccine candidates (targeting AAF/II, AAF/III, CS22, and BfpA) are deployed where clinical need is greatest.

4.8 Clinical Microbiology and Vaccine Pipeline Implications

These findings have immediate translational implications for diagnostic assay manufacturers, national essential diagnostics lists (NEDLs), and multivalent vaccine developers: 1. **Redesign of Molecular Diagnostic Panels:** First-tier multiplex PCR and qPCR/TAC panels that report “EAEC positive” based solely on `aatA`/`aggR` or “aEPEC positive” based solely on `eae+ bfpA-` suffer from poor positive predictive value for acute diarrhoea in endemic LMICs. Diagnostic assays should incorporate second-tier virulence markers: **`sepA`**, **`aafA` (AAF/II)**, **`agg3A`**

(AAF/III), **CS22**, and **pet**/**sat** for EAEC; **bfpA** and the OI-122 triad (**lifA/efa1**, **nleB**, **nleE**) for EPEC; and **sat** plus **afaE** for DAEC. 2. **Age-Stratified Clinical Interpretation of DAEC:** Detection of daaC/afaBC-positive DAEC in infants <12 months should be interpreted cautiously as likely asymptomatic colonization (OR $=0.91$), whereas detection in children aged $\geq 12-59$ months represents a clinically significant enteropathogen (OR $=2.21$) warranting inclusion in paediatric diarrhoea burden estimates. 3. **Subunit Vaccine Target Prioritization:** Because canonical EAEC regulators (aggR) and dispersin (aap) are ubiquitous in commensal carriage, multivalent DEC vaccine pipelines targeting African and global paediatric populations should prioritize high-attribution surface adhesins and secreted toxoids—specifically **AAF/II (aafA)**, **AAF/III (agg3A)**, **CS22**, **SepA toxoid**, **BfpA**, and **ETEC LT/ST + CFs**—to prevent acute clinical episodes and environmental enteric dysfunction.

4.9 Methodological Strengths and Limitations

This study has several notable strengths: (i) strict adherence to PRISMA 2020 guidelines across complementary pathotype and virulome evidence tiers covering 18 African countries and five global continents; (ii) exact separation of unassayed pathotypes (NA) from zero counts, preventing artificial deflation of pooled pathotype prevalence; (iii) dual reporting of standard Wald z-test and conservative Hartung–Knapp–Sidik–Jonkman (KH) t-distribution confidence intervals; (iv) formal multivariate REML meta-regression partitioning between-study variance and eliminating residual inconsistency ($I^2_{\text{res}} = 0.0\,\%$); (v) rigorous GRADE certainty profile construction across 28 pathotypes and virulence loci; and (vi) near-zero residual heterogeneity ($I^2 = 0.0 - 1.4\,\%$) across all high-attribution virulence loci in the virulome analysis, confirming that gene-level and age-stratified resolution eliminates the heterogeneity observed in unstratified pathotype pooling.

Several limitations warrant consideration. First, Northern and Central Africa remain underrepresented ($k = 1$ and $k = 3$ primary African surveillance cohorts, respectively) compared with Western, Eastern, and Southern Africa, underscoring the need for expanded genomic surveillance in Francophone Central and Saharan Africa. Second, while qPCR/TAC platforms provide cycle-threshold (C_t) quantification, most published case-control studies report binary presence/absence counts, precluding a continuous dose–response meta-regression of C_t values across all 29 virulence loci. Third, some individual accessory virulence loci (CS22, agg4A, afaE, nleB, nleE) were evaluated in $k = 3$ cohorts; although REML z-tests were highly significant and between-study heterogeneity was zero ($I^2 = 0.0 - 1.4\,\%$), the conservative $df = 2$ small-sample adjusted intervals were wide, highlighting the need for routine reporting of these loci in future WGS case-control studies.

4.10 Methodological Innovations, Resolution of Diagnostic Disparities, and Continental Evidence Synthesis

Earlier continental literature compilations, notably the preliminary synthesis by Kalule et al. (2024) (*Journal of Epidemiology and Global Health*, DOI: 10.1007/s44197-024-00301-w) which compiled 45 African studies ($N = 14,927$) published through December 2022, provided an important initial summary of raw African pathotype proportions and broad antimicrobial resistance trends. However, single-arm prevalence series inherently leave unresolved the foundational diagnostic, epidemiological, and genetic enigmas that have confounded the DEC

field for decades. By constructing a comprehensive, multi-platform diagnostic evaluation framework (**Supplementary Figure S17**) and executing an independent dual-track evidence synthesis, our study achieves decisive methodological and substantive advancements across five structural dimensions:

1. **Empirical Evidence Base and Global Comparative Scope:** Prior reviews were restricted exclusively to single-country African prevalence series ($k = 45$, $N = 14,927$, 80% hospital-based series). In contrast, our African surveillance synthesis expands the continental African evidence base through March 2026 ($N = 15,750$ across 18 countries), systematically capturing the decisive wave of post-2022 high-resolution genomic and microfluidic cohorts that appeared after earlier review cutoffs—including the multi-country Vaccine Impact on Diarrhea in Africa (VIDA) study (Ochieng et al., 2023), the multi-state Nigerian WGS cohorts (Akinlabi et al., 2023a, 2023b, 2025; Olayinka et al., 2024; Medugu et al., 2025), high-throughput Ethiopian surveillance (Zelelie et al., 2023; Wolde et al., 2024), and Zambian real-time PCR arrays (Mwape et al., 2023). Crucially, our study bridges continental African surveillance with a parallel global case-control virulome meta-analysis (Virulome synthesis: $k = 30$ cohort entries, 154 locus-level 2×2 evaluations across Africa, Latin America, Europe, Asia, and Oceania), achieving an unprecedented combined evidence base that establishes this work as the largest DEC molecular synthesis globally.

2. **Resolving the EAEC and DAEC Attribution Paradox via Matched Case-Control Modeling:** Earlier literature syntheses evaluated only single-arm case proportions, reporting high pooled case prevalence for EAEC (43%), ETEC (36%), DAEC (36%), and EPEC (31%). However, because healthy community controls were not incorporated, single-arm studies were unable to distinguish true pathogenic causality from benign commensal carriage, inherently assuming that all detected DEC isolates caused acute diarrhoea. This methodological limitation obscured the central paradox of enteric microbiology: in modern community surveillance, EAEC and DAEC are carried at equal or higher rates in asymptomatic healthy children. Our study directly resolves this paradox by synthesizing matched case-control Odds Ratios across 13 multi-centre African cohorts ($N = 10,724$) and international cohorts (**Figure 3**, **Supplementary Figure S3**), proving that unstratified EAEC (OR = 1.38) and DAEC (OR = 1.21) exhibit widespread non-pathogenic colonization, whereas ETEC (OR = 2.37), typical EPEC (OR = 2.42), and EIEC (OR = 2.43) are uniform acute disease drivers.

3. **High-Resolution Virulence Locus Meta-Analysis (29 Genetic Determinants vs. Coarse Pathotypes):** A critical limitation of earlier syntheses was the complete absence of individual virulence gene analysis; pathotypes were treated as coarse, monolithic entities. This coarse aggregation prevented them from explaining why some EAEC or DAEC strains cause life-threatening diarrhoea while others reside innocuously in the healthy gut. Our locus-level synthesis provides the first systematic meta-analysis of 29 distinct virulence loci across five continents (**Figure 3**, **Supplementary Figure S10**), demonstrating that clinical pathogenicity within EAEC is driven by specific tissue-damaging serine protease autotransporters (`sepA`, OR = 4.68; `pet`, OR = 2.19; `sat`, OR = 1.78) and high-affinity fimbrial adhesins (`aafA`, OR = 3.08; `agg3A`, OR = 2.71; `CS22`, OR = 3.59). Conversely, canonical regulon markers (`aggR`, `aatA`, `aap`), the

SPATE mucinase `pic` (showing balanced 16–22% carriage across cases and controls), and iron-acquisition siderophores (`fyuA`, `irp2`) are ubiquitous in healthy community controls, directly explaining why conventional single-gene PCR assays suffer devastating positive predictive value failure.

4. **Multi-Pathotype Genomic Hybrid Mobilome Architecture and Platform Discordance:** Prior reviews omitted hybrid pathotypes from their primary analysis, relegating them to an uncharacterized marginal footnote without mobilome context. In contrast, our study delineates the first multi-pathotype overlap network across Africa (**Figure 2**, **Supplementary Figure S3**, **Supplementary Figure S14**), demonstrating that `EAEC/ETEC` (52.3%), `STEC/EAEC` (24.1%), `EPEC/ETEC` (13.8%), and `DEC/ExPEC` (9.8%) co-hybrids represent a substantial public health threat driven by conjugative plasmid-phage synergy (*pAA*, *pEnt*, *LEE*, and lambdoid prophages). Furthermore, our platform stratification reveals that hybrid detection jumps from 5.5% under conventional endpoint PCR to 13.9% under WGS, exposing a 22.2% endpoint PCR false-negative discordance rate (**Figure 2**, **Supplementary Figure S7, S17**).

5. **Advanced Statistical Modeling, Variance Partitioning, and Evidence Certainty:** Where earlier compilations relied on basic unweighted or unadjusted pooling without multivariable meta-regression or evidence grading, and reported zero antimicrobial susceptibility data for African DAEC, our study deploys Freeman–Tukey double-arcsine transformations (`PFT`), mixed-effects multivariate meta-regression that accounts for 100% of residual between-study variance ($R^2_{\text{model}} = 100\%$, $I^2_{\text{res}} = 0.0\%$), Newcastle–Ottawa Scale and Hoy et al. risk-of-bias dashboards, formal GRADE certainty profiling across 28 pathotypes and loci (**Table 4**, **Supplementary Figure S16**), and complete antimicrobial susceptibility profiling across all six DEC pathotypes—including the first resolved African DAEC resistance profile ($n = 80$, showing 91.3% ampicillin and 82.5% trimethoprim-sulfamethoxazole resistance; **Supplementary Figure S8**).

Together, these methodological advances establish our synthesis as the definitive reference standard for DEC molecular epidemiology, diagnostic assay development, and pediatric vaccine design.

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5. Conclusions

Across Africa and globally, the diarrhoeagenic *Escherichia coli* attribution paradox is an artefact of pathotype-level diagnostic aggregation rather than true clinical non-pathogenicity. While EIEC (OR = 2.43), typical EPEC (OR = 2.42), and ETEC (OR = 2.37) act as uniform acute pathogens at the pathotype level, the high burden of EAEC (28.3% case prevalence), atypical EPEC (10.9%), and DAEC (4.9%) comprises a mixture of widespread endemic colonization and hypervirulent clinical clones. Resolving this paradox requires transitioning clinical microbiology and global surveillance workflows from single-marker pathotype assignment (`aatA`/`aggR`, unstratified `eae`, or age-unstratified `daaC`) to locus-resolved and age-stratified attribution targeting `sepA` (OR = 4.68), `CS22` (OR = 3.59), `aafA` (OR = 3.08), `agg3A` (OR = 2.71), `pet` (OR = 2.19), `sat` (OR = 1.78), and `aaiC` (OR = 1.47) in EAEC; children aged ≥ 12 – 59 months (OR = 2.21), `sat` (OR = 3.22), and `afaE` (OR = 3.07) in DAEC; `bfpA` (OR = 2.82) and the OI-122

effectors `lifA/efa1` (OR = 6.01), `nleB` (OR = 4.20), and `nleE` (OR = 3.80) in EPEC; and whole-genome or multi-target screening capable of capturing two-way hybrid DEC lineages (13.9 % by WGS).

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